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# Identifiability-Aware Source Attribution of Air Pollution

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## Abstract

Source attribution of urban  $\text{PM}_{2.5}$  from a monitoring network can be posed as a non-negative linear inverse problem: proxy emission maps and a wind-driven transport model give the concentration that each source group produces at each monitor and hour, and the readings are fitted as a nonnegative combination of these signals plus background terms. A fit with a small residual need not determine the contributions: sources aligned with the wind can leave nearly proportional signals at the sensors, and terms for regional background can absorb a source’s signal. We present Identifiability-Aware Source Attribution (IASA), which determines from the transport operator, before fitting, which groups of sources the readings can separate, and reports each group’s contribution with an error bound. A grouping is identifiable exactly when the column spaces of its blocks form a direct sum, and the finest such grouping is unique and computable in polynomial time. Moreover, the error of a group’s fitted signal is at most the noise in the operator’s column space divided by the group’s separation from the other sources. In controlled experiments on the New Delhi domain, IASA flags an absorbed source and a nearly proportional pair that the baselines do not flag, with group-signal errors of 0.4–6.0% (0.4–6.2% for nonnegative least squares after background removal, 36–784% for a mass balance without it). On four observed weeks, the four source groups are separable in every week, and at the estimated instrument noise level the error bound of the largest group’s signal is below its fitted norm in two of the four weeks.

## 1 INTRODUCTION

Air-quality policy in cities such as New Delhi needs to know which sources produce the  $\text{PM}_{2.5}$  measured at the monitors, and published source apportionments for the same Indian city disagree (Pant and Harrison, 2012; Karagulian et al., 2015). Receptor models estimate contributions from the chemical composition of particulate samples (Watson et al., 2002; Paatero and Tapper, 1994), whereas we combine the hourly  $\text{PM}_{2.5}$  and wind records of the New Delhi regulatory network (CPCB, 2023, 2025) with proxy emission maps for a few declared source groups (brick kilns, industries, population-related activity, traffic) and a transport model driven by the observed wind.

The readings then follow a nonnegative linear model whose columns are the  $\text{PM}_{2.5}$  that one unit of a source group’s activity produces at every sensor and hour, plus a low-rank nuisance term for regional background and smooth trends (Section 3).

A fit of this model with a small residual does not determine the contributions: a strong distant source and a weaker nearby source on the same wind line produce nearly proportional signals at the sensors, the nuisance term can absorb a source’s signal, and a sparse network may not see a source at all. Very different splits then fit the readings equally well, and a least-squares fit reports one regardless. In our controlled experiments, a fit without background removal reports group contributions with errors of 36–784% (Section 6).

We therefore compute, before fitting, which groups of sources the network can separate, and report each group’s contribution and share only at that resolution.

### Contributions.

- We formulate  $\text{PM}_{2.5}$  attribution from a regulatory network as a nonnegative linear inverse problem (Section 3) and give a procedure, Identifiability-Aware Source Attribution (IASA), that fixes the reported grouping before fitting and reports group signals, shares, error bounds and intervals (Section 4).
- We show that a grouping of sources is identifiable

if and only if the column spaces of its blocks form a direct sum, and that the finest such grouping is unique and computable in  $O(NJ^2)$  operations (Theorem 5.7). We also show that the error of a group’s fitted signal is at most the noise in the operator’s column space divided by the group’s separation from the other sources (Theorem 5.9), and we give a condition under which the set of reportable groupings is unchanged by an error in the wind-driven operator (Proposition 5.12, Section 5).

- In controlled experiments with known contributions on the New Delhi domain, IASA flags an absorbed source and a pair of nearly proportional sources that no baseline flags, with group-signal errors of 0.4–6.0% (plain NNLS after background removal: 0.4–6.2%, no flag) (Section 6.1). On four observed weeks from the 32-sensor network the four source groups are separable in every week, but at the instrument noise level the error bound of the largest group’s signal is below its fitted norm in only two of the four weeks (Section 6.2).

## 2 RELATED WORK

**Receptor models.** Chemical mass balance (CMB) and positive matrix factorization (PMF) estimate source contributions from the measured chemical composition of particulate samples (Watson et al., 2002; Paatero and Tapper, 1994; Hopke, 2016), and reviews document their sensitivity to source profiles and collinear tracers (Viana et al., 2008; Belis et al., 2013, 2019; Mircea et al., 2020; Hopke et al., 2020). Studies of Indian cities report divergent contributions for the same city (Pant and Harrison, 2012; Karagulian et al., 2015). Henry (1992) defines an eligible space of estimable combinations for CMB with nearly collinear profiles, PMF rotational ambiguity is explored with dedicated tools (Paatero et al., 2002), and statistical treatments address temporal correlation (Park et al., 2001) and identification (Jin and Datta, 2025). Our analysis uses the  $\text{PM}_{2.5}$  and wind records of the New Delhi network, so the sources are separated by transport and timing rather than by particle composition, and the network’s gas measurements are not used.

**Source-oriented models and inversion.** Inverse modelling and data assimilation estimate emissions from concentrations with a prior (Tarantola, 2005; Carrassi et al., 2018), and atmospheric inversion measures what the observations constrain through the averaging kernel (Rodgers, 2000) and chooses how far to aggregate the state by balancing aggregation and smoothing error (Turner and Jacob, 2015). IASA also aggregates, but without a prior: it reports the reportable grouping of declared sources with the most groups. Spatio-temporal

models for sensor networks (Li et al., 2017; Yu et al., 2017; Iyer et al., 2022; Bhardwaj et al., 2025a) target forecasting and interpolation rather than attribution.

**Estimability and collinearity.** Estimability (Searle, 1971), identification (Rothenberg, 1971; Manski, 2003) and collinearity diagnostics (Fox and Monette, 1992; Belsley et al., 1980) do not return a partition. The partition of Theorem 5.7 is the component decomposition of the column matroid (Kroghdahl, 1977; Oxley, 2011), and the separation is the sine of a principal angle (Björck and Golub, 1973). Appendix H gives a longer discussion.

## 3 PROBLEM SETUP

**Network and records.** The Central Pollution Control Board publishes hourly  $\text{PM}_{2.5}$  and wind for the New Delhi regulatory monitors (CPCB, 2023, 2025). We use the records from 1 May 2018 to 31 October 2020 at 32 locations, with the two co-located Pusa monitors averaged and no readings at East Arjun Nagar. At the 31 locations with readings, 93.4% of hourly  $\text{PM}_{2.5}$ , 77.3% of wind-direction and 75.1% of wind-speed entries are present, and three locations record no wind.  $\text{PM}_{2.5}$  is never imputed. Wind is converted from the direction it comes from to the direction of transport and interpolated with a normalized Gaussian kernel onto a  $40 \times 40$  grid that spans the bounding box of the sensor locations (Appendix D).

**Source groups.** Four source groups are declared from proxy maps (Figure 1): brick kilns and industries from a regional emissions inventory (Guttikunda and Calori, 2013), population density from a gridded population product (Center for International Earth Science Information Network (CIESIN), Columbia University, 2018), and traffic from a road-network map (Google, 2024). Each inventory map is cropped to a fixed  $40 \times 40$  window of its  $80 \times 80$  grid, whose alignment with the sensor bounding box is not verified (Appendix D.6), and divided by its own 99th percentile, and the traffic map is the mean of four time-of-day road maps normalized this way (the 06:00 map in the inventory is zero). The activity of group  $k$  is  $\theta_k(t) = \sum_b c_{kb} \phi_b(t)$  with  $c_{kb} \geq 0$  and declared nonnegative temporal basis functions  $\phi_b$ : alternating 12-hour blocks for brick kilns, one component for industries with a continuous level and higher daytime (07–19 h) activity, peaks at 07, 13 and 19 h for population, and time-of-day slots at 00, 06, 12 and 18 h for traffic. A component is a pair  $j = (k, b)$ . Each group uses only its own basis functions, so the coefficients of the others, the set  $\mathcal{F}_0$ , are set to zero before fitting, and no coefficient is removed after fitting because its estimate is small.  $J$  is the number of remaining components. Because each proxy map has its

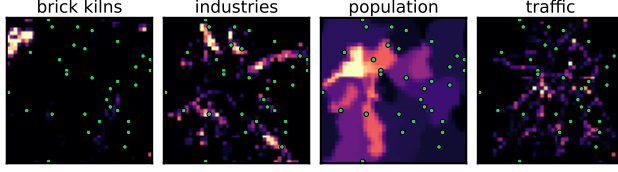


Figure 1: Proxy maps of the four source groups on the  $40 \times 40$  grid (each inventory map divided by its own 99th percentile, with colour clipped at 1) and the 32 regulatory sensor cells (green). Traffic is the mean of the four time-of-day road maps.

own normalization, the coefficients are in normalized proxy units, whereas the group signals defined below are in  $\mu\text{g}/\text{m}^3$  and do not depend on that normalization.

**Transport operator.** Emissions released at hour  $t - \ell$  reach the sensors at hour  $t$  after transport by the wind between the two hours. Let  $G_{t,t-\ell}$  map emissions on the source grid at  $t - \ell$  to concentrations at the  $m$  sensors at  $t$  under an open-boundary Gaussian-puff model driven by the gridded wind  $\omega$  (mass that leaves the domain is removed), and let  $\mathbf{s}_k$  be the proxy map of group  $k$ . The  $\text{PM}_{2.5}$  that one unit of  $c_{kb}$  produces at the sensors at hour  $t$  is

$$\mathbf{a}_{t,kb}(\omega) = \sum_{\ell=0}^{\min(L,t-1)} \phi_b(t-\ell) G_{t,t-\ell} \mathbf{s}_k \in \mathbb{R}^m. \quad (1)$$

In  $G_{t,t-\ell}$ , a unit release is advected along the interpolated wind and spread by a Gaussian kernel, evaluated at the sensors, whose variances along and across the mean wind grow as  $0.7^2$  and  $0.25^2$  square cells per hour of travel. The model is two-dimensional and treats  $\text{PM}_{2.5}$  as a passive primary tracer, without mixing height, deposition or secondary formation (Appendix D.3). Stacking over hours gives column  $\mathbf{a}_{kb}(\omega)$  of the operator  $A(\omega)$ . The lag is declared before fitting:  $L = 12$  hours in every reported run except the lag sweep of Experiment 7.

**Model, nuisance and projection.** Let  $\mathbf{y} \in \mathbb{R}^N$  stack the observed hourly readings; rows with a missing reading are removed from  $\mathbf{y}$ ,  $A(\omega)$  and  $Q$  alike. The model is

$$\mathbf{y} = A(\omega) \mathbf{c} + Q\boldsymbol{\gamma} + \boldsymbol{\epsilon}, \quad \mathbf{c} \geq 0, \quad (2)$$

with  $\mathbf{c} \in \mathbb{R}_+^J$ , free nuisance coefficients  $\boldsymbol{\gamma}$  and noise  $\boldsymbol{\epsilon}$ . The nuisance basis  $Q \in \mathbb{R}^{N \times r}$  represents regional background and smooth trends. Except in the nuisance-basis variations of Experiments 3 and 11, it has rank  $r = 4$ : a constant, a linear time trend and one daily sine-cosine pair, declared before fitting and never built from the inventories or from fitted signals. With  $P_Q^\perp = I - QQ^\dagger$ , the projected problem is

$$\tilde{\mathbf{y}} = \tilde{A}\mathbf{c} + \tilde{\boldsymbol{\epsilon}}, \quad \tilde{\mathbf{y}} = P_Q^\perp \mathbf{y}, \quad \tilde{A} = P_Q^\perp A(\omega). \quad (3)$$

The projected column  $\tilde{\mathbf{a}}_j = P_Q^\perp \mathbf{a}_j$ , the *fingerprint* of component  $j$ , is the part of its signal that the nuisance cannot reproduce, and because the projection can absorb a signal, every diagnostic is computed from  $\tilde{A}$ .

**The attribution problem.** Let  $E = \{1, \dots, J\}$  index the components and let  $\mathcal{D}$  be the declared partition of  $E$  into source groups. For  $g \subseteq E$ ,  $\tilde{A}_g$  is the submatrix of the columns in  $g$ , and the *signal* of  $g$  is  $\tilde{A}_g \mathbf{c}_g$ , its contribution to the projected readings. It is in  $\mu\text{g}/\text{m}^3$  and does not change when a proxy map is rescaled; a sum of coefficients  $\sum_{j \in g} c_j$  has neither property. The task is to choose a partition  $\mathcal{P}$  of  $E$  whose blocks are unions of blocks of  $\mathcal{D}$ , using only  $\tilde{A}$ ,  $\mathcal{D}$  and a declared noise level, before the readings are fitted, and then to report for each block  $g$  its fitted signal  $\tilde{A}_g \hat{\mathbf{c}}_g$ , its share  $\|\tilde{A}_g \hat{\mathbf{c}}_g\|_1 / \sum_{h \in \mathcal{P}} \|\tilde{A}_h \hat{\mathbf{c}}_h\|_1$ , and the uncertainty of both. The partition must meet two requirements. (R1) *Identifiability*: without noise, the readings determine the signal of every block. (R2) *Reportability*: at the declared noise level, the error of every block's fitted signal is bounded by a stated multiple of the noise. R1 excludes splits that no amount of data resolves, while R2 excludes splits that the noise does not allow. Section 4 gives the procedure and Section 5 the conditions under which it meets both.

## 4 METHOD: IDENTIFIABILITY-AWARE SOURCE ATTRIBUTION

IASA computes from  $\tilde{A}$ , before fitting, diagnostics and the partition to report, then fits once (Algorithm 1, with thresholds in Table 3). Contributions are estimated by nonnegative least squares (NNLS),

$$\hat{\mathbf{c}} = \arg \min_{\mathbf{c} \geq 0} \|\tilde{\mathbf{y}} - \tilde{A}\mathbf{c}\|_2^2, \quad (4)$$

solved by projected FISTA (Beck and Teboulle, 2009) (Appendix K), and it is the same estimator in every experiment.

**Diagnostics.** IASA reports the rank of  $\tilde{A}$ , its smallest singular value  $\sigma_J$  (zero if rank deficient),  $\kappa = \sigma_1/\sigma_J$ , the visibility  $\|\tilde{\mathbf{a}}_j\|_2$ , the weak set of columns with visibility at most  $\tau_v$  ( $\tau_v = 0$  in every run), the absorption  $\|P_Q \mathbf{a}_j\|_2 / \|\mathbf{a}_j\|_2$ , which is the fraction of a column that the projection removes, and the coherence  $\rho_{ij} = |\tilde{\mathbf{a}}_i^\top \tilde{\mathbf{a}}_j| / (\|\tilde{\mathbf{a}}_i\|_2 \|\tilde{\mathbf{a}}_j\|_2)$ . The central diagnostic is the *separation* of a group  $g \subseteq E$ ,

$$s_g = \sin \theta_g, \quad (5)$$

where  $\theta_g$  is the smallest principal angle (Björck and Golub, 1973) between  $\text{col}(\tilde{A}_g)$  and  $\text{col}(\tilde{A}_{E \setminus g})$ , with

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**Algorithm 1** Identifiability-Aware Source Attribution (IASA)
 

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- 1: **Declare**  $Q$ , components,  $\mathcal{D}$ ,  $\mathcal{F}_0$ ,  $L$ ,  $\sigma_e$ ,  $\delta$  and thresholds.
  - 2: **Operator:** build  $A(\omega)$  by (1).
  - 3: **Project:** drop rows with a missing reading and form  $\tilde{\mathbf{y}}$ ,  $\tilde{A}$  (3).
  - 4: **Diagnose:** singular values, rank, visibility, absorption, coherence, weak set.
  - 5: **Group:**  $\mathcal{P}^*$ , then atoms  $\mathcal{P}^* \vee \mathcal{D}$ , then the reportable partition at  $\tau_\theta$  with the most blocks.
  - 6: **Fit:** solve (4) and form group signals  $\tilde{A}_g \hat{\mathbf{c}}_g$  and shares.
  - 7: **Uncertainty:** error bound  $b_g$  of each block at  $\sigma_e$ , bootstrap intervals of the shares, and a residual adequacy test if a calibrated noise model exists.
  - 8: **Report** group signals, shares,  $b_g$ , intervals, coefficients of single-component blocks, diagnostics and flags.
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$s_g = 1$  when either space is  $\{0\}$ . In words,  $s_g$  near 0 means that some signal of group  $g$  is nearly reproduced by the other sources. For an operator with two columns  $s = \sqrt{1 - \rho^2}$ , and for one column  $1/s_J^2$  is the uncentered variance-inflation factor. IASA raises an identifiability *flag* on a merge, a weak column, an absorption at least 0.9,  $\sigma_J \leq 10^{-6}$  or a coherence at least 0.99.

**Grouping.** The partition is chosen in two steps. First, IASA computes the finest exactly identifiable partition  $\mathcal{P}^*$  of the columns (Theorem 5.7(c)), and the blocks of  $\mathcal{P}^* \vee \mathcal{D}$ , the finest partition coarser than both, are the *atoms*. Second, a partition into unions of atoms is *reportable* if every block has  $s_g \geq \tau_\theta$ . IASA returns the reportable partition with the most blocks, breaks ties by the largest smallest separation, and lists any remaining ties. With  $q \leq 12$  atoms this search is exact by dynamic programming in  $O(3^q)$  time, whereas with more atoms it merges greedily (Appendix A). We use  $\tau_\theta = \sqrt{1 - 0.99^2} = 0.141$ , so for an operator with two columns the rule merges exactly when their coherence exceeds 0.99. The recorded runs used either this rule or a pairwise rule that merges declared blocks with a cross-source coherence above 0.99, and on every saved operator and observed week the two rules give the same partition.

**Uncertainty and adequacy.** With a declared noise standard deviation  $\sigma_e$  per reading, IASA gives for each reported block the bound  $b_g = \sigma_e(\sqrt{\text{rank}(\tilde{A})} + \sqrt{2 \log(1/\delta)})/s_g$  on the error of its fitted signal, which holds with probability at least  $1 - \delta$  under the conditions of Theorem 5.9(c) ( $\delta = 0.05$  here). We call a group's signal *resolved* when  $b_g$  is below the norm of its fitted signal. It also gives percentile intervals of the shares from a parametric bootstrap: since the NNLS estimate depends on the readings only through  $P_{\tilde{A}} \tilde{\mathbf{y}}$ , with  $P_{\tilde{A}}$  the orthogonal projector onto  $\text{col}(\tilde{A})$ , the bootstrap adds Gaussian noise of standard deviation  $\sigma_e$  in

$\text{col}(\tilde{A})$  to  $\tilde{A} \hat{\mathbf{c}}$  and refits. A residual adequacy test is run when a calibrated noise model exists, and otherwise the residual is labelled uncalibrated (Appendix C).

## 5 THEORETICAL GUARANTEES

This section gives the guarantees behind steps 2, 4, 5 and 7 of Algorithm 1, and the proofs are in Appendix A. Steps 1–3 construct  $\tilde{A}$  before any reading is fitted, and the analysis treats  $\tilde{A}$  as fixed, which requires the following.

**Assumption 5.1** (Declared model).  $\mathbf{y}$  follows (2) with  $\mathbf{c} \geq 0$ , where  $\mathbf{y}$  is the readings minus a declared baseline field, and  $A(\omega)$ ,  $Q$ , the components,  $\mathcal{F}_0$  and the set of observed rows are fixed independently of  $\epsilon$ .

In the observed runs the baseline is the transported field of a map interpolated from the first hour of readings, and its error enters  $\epsilon$ . The assumption fails if  $Q$  or the components are chosen after looking at estimates, or if the transport model is wrong (Section 5.4). Step 7 also requires a noise model.

**Assumption 5.2** (Noise).  $\epsilon \sim N(0, \sigma^2 I_N)$  (or, where stated,  $N(0, \Sigma)$ ), and the declared level satisfies  $\sigma_e \geq \sigma$  (or  $\sigma_e^2 \geq \|\Sigma\|_2$ ).

On the observed data the difference of the two co-located monitors has lag-1 autocorrelation 0.857, so independence in time does not hold, and  $\sigma_e \geq \sigma$  is not checked (Section 6.2).

### 5.1 Individual Coefficients

We begin with the individual source-basis coefficients, such as the four time-of-day activities of traffic.

**Proposition 5.3** (Exact identifiability). *In the noiseless projected model  $\tilde{\mathbf{y}} = \tilde{A} \mathbf{c}$ ,  $\tilde{A} \mathbf{c}_1 = \tilde{A} \mathbf{c}_2$  implies  $\mathbf{c}_1 = \mathbf{c}_2$  for all  $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}_+^J$  if and only if  $\text{rank}(\tilde{A}) = J$ .*

**Proposition 5.4** (Noise robustness). *Assume  $\text{rank}(\tilde{A}) = J$  and let  $\sigma_J(\tilde{A})$  be the smallest singular value. If  $\hat{\mathbf{c}}$  is the least-squares estimate, or the nonnegative least-squares (NNLS) estimate and  $\mathbf{c} \geq 0$ , then*

$$\|\hat{\mathbf{c}} - \mathbf{c}\|_2 \leq \|P_{\tilde{A}} \tilde{\epsilon}\|_2 / \sigma_J(\tilde{A}) \leq \|\tilde{\epsilon}\|_2 / \sigma_J(\tilde{A}). \quad (6)$$

*Remark 5.5* (Implication: coefficients). Both propositions require only Assumption 5.1. In every observed week  $\text{rank}(\tilde{A}) = 7 = J$ . With noise,  $1/\sigma_J$  is the largest factor by which noise enters the coefficients, so an identifiable problem can still have large coefficient errors (Experiment 1). IASA reports  $\sigma_J$  in step 4 and reports coefficients only for single-component blocks.

## 5.2 Identifiable Groupings

When the coefficients are not determined, the reported quantity is the signal of a group of sources. The following definition states when such signals are determined, which is requirement R1.

**Definition 5.6** (Identifiable grouping). A partition  $\mathcal{P}$  of  $E$  is *identifiable* if  $\tilde{A}\mathbf{c}_1 = \tilde{A}\mathbf{c}_2$  with  $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}_+^J$  implies  $\tilde{A}_g\mathbf{c}_{1,g} = \tilde{A}_g\mathbf{c}_{2,g}$  for every block  $g \in \mathcal{P}$ .

**Theorem 5.7** (Identifiable groupings).

- (a)  $\mathcal{P}$  is identifiable if and only if  $\text{col}(\tilde{A}) = \bigoplus_{g \in \mathcal{P}} \text{col}(\tilde{A}_g)$ , equivalently  $\text{rank}(\tilde{A}) = \sum_{g \in \mathcal{P}} \text{rank}(\tilde{A}_g)$ .
- (b) There is a unique finest identifiable partition  $\mathcal{P}^*$ , and  $\mathcal{P}$  is identifiable if and only if every block of  $\mathcal{P}$  is a union of blocks of  $\mathcal{P}^*$ .
- (c) Let  $B \subseteq E$  index a basis of  $\text{col}(\tilde{A})$  chosen from the columns, and join each  $e \notin B$  to every  $b \in B$  whose coefficient is nonzero when  $\tilde{\mathbf{a}}_e$  is written in that basis. The blocks of  $\mathcal{P}^*$  are the connected components of this graph, for any choice of  $B$ , and they are the connected components of the column matroid of  $\tilde{A}$ .
- (d) For a declared partition  $\mathcal{D}$ , the finest identifiable partition whose blocks are unions of blocks of  $\mathcal{D}$  is the finest common coarsening  $\mathcal{P}^* \vee \mathcal{D}$ .

**Remark 5.8** (Implication: exact resolution). The theorem requires only Assumption 5.1. A partition meets R1 exactly when no group's span meets the span of the other groups except at 0 (a), and there is one finest such partition (b), computed before fitting in  $O(NJ^2)$  arithmetic (c). A source shares its block of  $\mathcal{P}^* \vee \mathcal{D}$  with another source exactly when a nonzero combination of its fingerprints equals a combination of the other fingerprints. In every observed week  $\mathcal{P}^*$  is the seven single columns, so the atoms are the four groups. Step 5 forms its atoms by (c) and (d), so every partition IASA can return meets R1. For sums of coefficients, which are in different proxy units, the minimal groupings can be non-unique and deciding whether one with at least two blocks exists is NP-complete (Proposition A.1, Appendix A), so IASA reports signals instead.

## 5.3 Group-Signal Error

Requirement R2 concerns the effect of noise, which identifiability does not control. The error of a group's fitted signal is governed by its separation  $s_g$  of (5), not by  $\sigma_J$ .

**Theorem 5.9** (Group-signal error). Let  $\mathcal{P}$  be identifiable and let  $\hat{\mathbf{c}}$  be any least-squares solution, or any NNLS solution with  $\mathbf{c} \geq 0$ .

- (a) The group signals  $\tilde{A}_g\hat{\mathbf{c}}_g$  do not depend on which solution is chosen, and for every  $g \in \mathcal{P}$ ,

$$\|\tilde{A}_g(\hat{\mathbf{c}}_g - \mathbf{c}_g)\|_2 \leq \frac{\|P_{\tilde{A}}\tilde{\boldsymbol{\epsilon}}\|_2}{s_g}. \quad (7)$$

- (b) If  $\boldsymbol{\epsilon} \sim N(0, \sigma^2 I_N)$ , for a least-squares solution and  $r_g = \text{rank}(\tilde{A}_g)$ ,  $\mathbb{E}\|\tilde{A}_g(\hat{\mathbf{c}}_g - \mathbf{c}_g)\|_2^2 \leq \sigma^2 r_g / s_g^2$ , with equality when  $r_g = 1$ .
- (c) If  $\boldsymbol{\epsilon} \sim N(0, \sigma^2 I_N)$ , with probability at least  $1 - \delta$ , simultaneously for every identifiable partition and every block  $g$ ,

$$\|\tilde{A}_g(\hat{\mathbf{c}}_g - \mathbf{c}_g)\|_2 \leq \frac{\sigma(\sqrt{\text{rank}(\tilde{A})} + \sqrt{2\log(1/\delta)})}{s_g}, \quad (8)$$

and if  $\boldsymbol{\epsilon} \sim N(0, \Sigma)$ , the same holds with  $\sigma^2$  replaced by  $\|\Sigma\|_2$ .

**Remark 5.10** (Implication: the error bound IASA reports). Part (a) holds for every noise vector under Assumption 5.1, whereas (b) and (c) also need the Gaussian noise of Assumption 5.2. IASA, which uses NNLS, relies on (a) and (c). The columns inside a group, such as the four time-of-day columns of traffic, can be nearly dependent, which makes the bound of Proposition 5.4 large while the group's signal error stays at most  $\|P_{\tilde{A}}\tilde{\boldsymbol{\epsilon}}\|_2/s_g$ . With  $\sigma$  replaced by the declared  $\sigma_e$ , the right side of (8) is the bound  $b_g$  of step 7, which holds when  $\sigma_e \geq \sigma$  and the operator is exact. Requiring  $b_g \leq \eta$  for a tolerance  $\eta$  gives the threshold  $\tau_\theta = \sigma_e(\sqrt{\text{rank}(\tilde{A})} + \sqrt{2\log(1/\delta)})/\eta$  of step 5, and 0.141 corresponds to  $\eta = 7.1 \sigma_e(\sqrt{\text{rank}(\tilde{A})} + \sqrt{2\log(1/\delta)})$ .

**Definition 5.11** (Reportable partition). A partition into unions of atoms is *reportable at threshold*  $\tau_\theta \in (0, 1]$  if every block has  $s_g \geq \tau_\theta$ .

Reportable partitions are not closed under coarsening, and the minimal ones (with no strictly finer reportable partition) need not be unique (Proposition A.2), so with up to 12 atoms IASA checks every union of atoms and lists ties. The one-block partition is always reportable.

## 5.4 Errors in the Operator

The operator is built from an estimated wind field and assumed dispersion parameters. The next result bounds how far an error  $\Delta$  in  $\tilde{A}$  moves the separations.

**Proposition 5.12** (Stability under operator error). Let  $\tilde{A}' = \tilde{A} + \Delta$ . For  $S \subseteq E$  with  $\tilde{A}_S$  of full column rank and  $\|\Delta_S\|_2 \leq \sigma_{\min}(\tilde{A}_S)/2$ , let  $\beta_S = \arcsin(\|\Delta_S\|_2/(\sigma_{\min}(\tilde{A}_S) - \|\Delta_S\|_2))$ . If this holds for  $S = g$  and  $S = E \setminus g$ , with  $g$  nonempty and proper, then  $|\theta_g(\tilde{A}') - \theta_g(\tilde{A})| \leq \beta_g + \beta_{E \setminus g}$ . If  $\tilde{A}$  has full column rank,

$\|\Delta\|_2 \leq \sigma_J(\tilde{A})/2$ , and  $|\theta_g(\tilde{A}) - \arcsin \tau_\theta| > \beta_g + \beta_{E \setminus g}$  for every nonempty proper union  $g$  of atoms, the reportable partitions of  $\tilde{A}'$  and  $\tilde{A}$  coincide, and so do the minimal reportable partitions.

*Remark 5.13* (Implication: wind and dispersion error). The proposition uses no noise assumption. Under a wind or dispersion error  $\Delta$  that meets the conditions, the set of reportable partitions is unchanged, and so is the reported partition when one reportable partition has the most blocks. In every observed week each union of atoms has a separation angle of at least  $45.4^\circ$ , against  $\arcsin \tau_\theta = 8.1^\circ$ , but whether the true  $\Delta$  meets the conditions is not known. The fitted signals do change: for least squares, an operator error acts as additional noise of size at most  $\|\Delta\|_2 \|\mathbf{c}\|_2$  (Proposition D.1, Appendix D.4), which Experiment 5 measures.

## 6 EXPERIMENTS

We evaluate IASA in controlled experiments with known contributions (Section 6.1) and on four observed weeks without ground truth (Section 6.2).

### 6.1 Controlled Experiments

The controlled experiments generate PM<sub>2.5</sub> readings from known nonnegative contributions with the puff model of Section 3 on the  $40 \times 40$  grid, so that the error of every reported quantity can be measured. Each run covers 72 hours, mostly with two source groups, and varies one factor at a time: wind, sensor layout, nuisance basis, noise, operator error, lag, temporal basis or inventory. Unless the factor is varied, the sensors are the 32 regulatory locations, with three downwind sensors in Experiment 10 and six random ones in the steady-wind case of Experiment 11. The sources are compact synthetic emitters except in Experiments 2, 6 and 8, which use the New Delhi maps, and the winds are synthetic except where real wind is stated (Appendices E–G). Coefficient error is  $\|\hat{\mathbf{c}} - \mathbf{c}\|_2 / \|\mathbf{c}\|_2$ , the group-signal error of a reported block  $G$  is  $\|\tilde{A}_G(\hat{\mathbf{c}}_G - \mathbf{c}_G)\|_2 / \|\tilde{A}_G \mathbf{c}_G\|_2$ , and noise is Gaussian with standard deviation stated as a fraction of the largest absolute noiseless source signal of that run.

**Baselines on two controlled degeneracies.** IASA is compared over three seeds with plain NNLS on the projected system (the same estimator without grouping and diagnostics), a transport-based mass balance (CMB: NNLS on the unprojected response, with transport columns in place of chemical profiles) and a non-negative matrix factorization of the sensor-by-hour readings (PMF/NMF) (Figure 2(a), Table 1). Every method is scored on IASA’s reported blocks, errors

Table 1: Experiment 11: largest relative group-signal error of the reported blocks (background stress in seeds 0, 1 and 2, and steady wind in seed 2, the only seed with signal at the sensors). Share errors: Table 14.

| method     | flag | background stress   | steady wind    |
|------------|------|---------------------|----------------|
| IASA       | yes  | 0.054, 0.028, 0.060 | 0.004 (merged) |
| plain NNLS | no   | 0.054, 0.028, 0.062 | 0.004          |
| CMB        | no   | 4.63, 2.10, 4.32    | 7.84           |

range over those blocks, and share error is the  $L_2$  distance between share vectors. *Source-like background* (32 regulatory sensors, two sources with two columns each): a declared nuisance column is proportional to one basis column of source A, whose norm falls from 0.081 to  $10^{-16}$  under projection. IASA flags it (absorption 1), the two sources have separation above 0.9999, and IASA’s group-signal errors are 0.4–6.0%, with share errors 0.003–0.007. Plain NNLS gives group-signal errors within 0.003 of IASA’s and raises no flag. CMB, without nuisance removal, has group-signal errors of 36–463% and share errors 0.12–0.19. *Steady wind* (one wind direction, six random sensors): in two seeds neither plume reaches the sensors (column norms below  $10^{-10}$ , against 0.062 and 0.067 in the third), and IASA flags  $\sigma_J \approx 0$ . In the third seed the two columns have coherence 0.99966 and separation  $0.026 < \tau_\theta$ , so no reportable partition has two blocks. IASA therefore merges them, and the merged signal has error 0.44%, against 784% for CMB. Reported separately, as plain NNLS reports them, the two sources’ signals have errors of 3.8% and 6.9%. PMF/NMF has share errors on the unprojected response of 1.15–1.19 (background stress) and 0.30 (steady wind), and no baseline raises a flag in either scenario.

**Conditioning and noise (Experiment 1).** For two full-rank geometries, coefficient error is zero without noise and grows linearly with it (one noise draw, scaled): 0.066, 0.132, 0.264 at noise 0.05, 0.10, 0.20 for  $\sigma_J = 3.27$ , against 0.028, 0.057, 0.113 for  $\sigma_J = 9.60$  (Figure 2(b), Table 6), consistent with Proposition 5.4.

**Sensor layout and wind (Experiment 4).** With the 32 regulatory locations,  $\sigma_J \geq 1.53$  under all six winds, and the maximum coherence falls from 0.975 under a constant wind to 0.129 and 0.156 under multi-directional and real New Delhi wind (Figure 2(c), Table 7). With eight randomly placed sensors and four of the six winds, source B’s operator column has norm  $5.2 \times 10^{-7}$  to  $1.7 \times 10^{-4}$ , against 7.6–18.6 with the regulatory layout, so  $\sigma_J \approx 0$  and the largest coefficient error of the sweep (0.573) is that source’s poorly determined coefficient ( $\sigma_J \approx 10^{-6}$ ). The layouts also differ in sensor count.

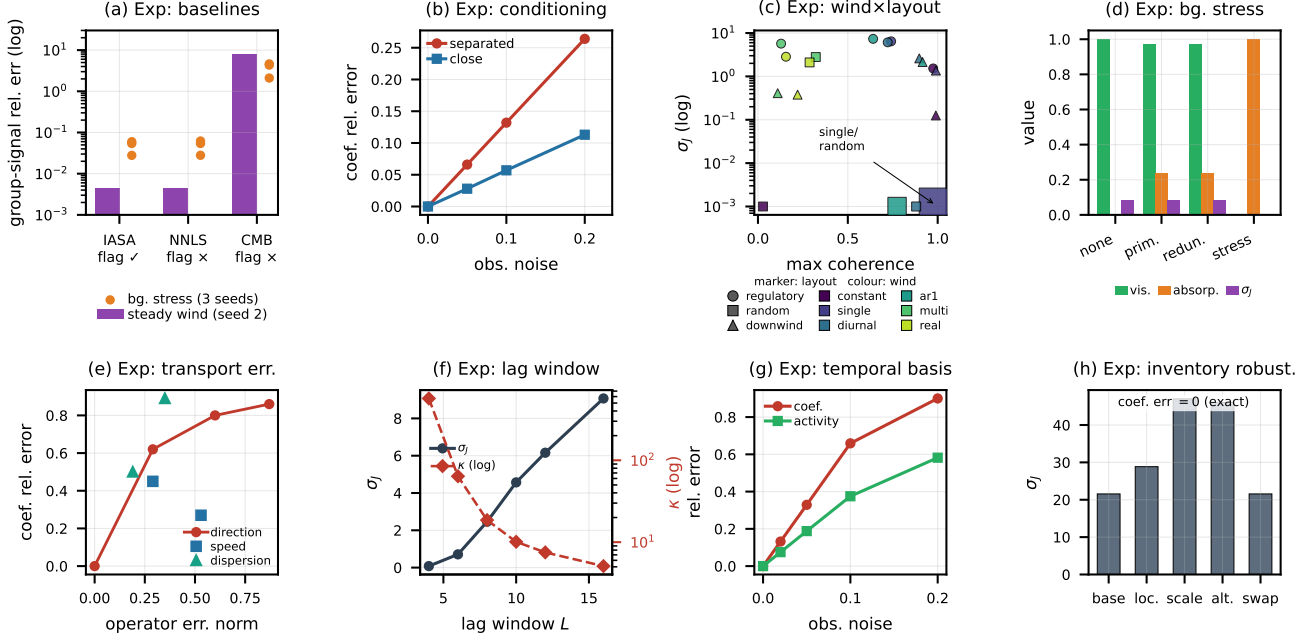


Figure 2: Controlled experiments on the New Delhi platform. **(a)** Largest relative group-signal error of IASA and two baselines that raise no identifiability flag (plain NNLS on the projected system, a mass balance on the unprojected response): background stress over three seeds (points) and the steady-wind case in the seed where the plumes reach the sensors (bars). **(b)** Coefficient error against noise for two sources 4 cells apart ( $\sigma_J = 9.6$ ,  $\kappa = 2.3$ ) and 20 cells apart ( $\sigma_J = 3.3$ ,  $\kappa = 8.9$ ). **(c)**  $\sigma_J$  against maximum coherence across wind providers (colour, with ar1 denoting AR(1) wind) and sensor layouts (marker), where marker size increases with coefficient error and  $\sigma_J$  below  $10^{-3}$  is drawn at  $10^{-3}$ . **(d)** Nuisance bases none, primary (rank 4), redundant (same span) and stress (source-like column): the stress basis drives absorption to 1 and  $\sigma_J$  and the visibility ratio  $\|\tilde{\mathbf{a}}_j\|_2/\|\mathbf{a}_j\|_2$  to 0. **(e)** Coefficient error against operator-error norm along the wind-direction, wind-speed and dispersion axes. **(f)**  $\sigma_J$  and  $\kappa$  against the lag window. **(g)** Activity error and coefficient error against noise. **(h)**  $\sigma_J$  across inventory versions.

**Nuisance absorption and missing sources (Experiments 3 and 8).** A source-like nuisance basis drives the absorption of one column to 1 and  $\sigma_J$  to 0 while the fitted residual is unchanged (Figure 2(d), Table 9), and the coefficient error falls (0.78 with the source-like basis against 2.61 with the rank-4 basis) because the signal has been absorbed, so the error alone would not reveal the problem. The adequacy test ( $\alpha = 0.05$ , 1,000 replicates) rejects at rate 0.05 under a correct model and 1.00 when an omitted source has 91% of its signal outside the span of  $A(\omega)$  and  $Q$ , whereas an omitted source equal to a nonnegative combination of the fitted columns plus nuisance columns is absorbed and rejected at rate 0.05 (Table 21). A non-rejection therefore does not show that the inventory is complete.

**Transport error (Experiment 5).** This experiment measures the operator error of Proposition D.1. Fitting with a perturbed operator (Figure 2(e), Table 10), the coefficient error reaches 0.62 at a  $5^\circ$  error in wind direction, 0.86 at  $20^\circ$  and 0.89 at twice the dispersion, while along the speed axis it is not monotonic (0.45 at

$1.25\times$ , 0.27 at  $1.5\times$ ). A structural mismatch (advection–diffusion data, puff operator) is rejected in 100% of trials.

**Lag, temporal basis and inventory (Experiments 7, 9 and 6).** From  $L = 4$  to 16 hours  $\sigma_J$  rises from 0.08 to 9.07, but the operator still changes by 6–14% between adjacent lags (Figure 2(f)). Activity trajectories have lower error than coefficients (0.58 against 0.90 at 20% noise, panel (g)), and  $\sigma_J$  moves from 21.6 to 47.0 under a rescaled inventory map while recovery stays exact (panel (h)).

## 6.2 Observed New Delhi Weeks

**Data.** We fit the first four weeks of the record, 1–28 May 2018, separately. Each uses the 79–85% of sensor-hours with a valid reading ( $N = 4,236$ – $4,561$ ) and  $J = 7$  columns after  $\mathcal{F}_0$ : one temporal component each for brick kilns, industries and population, and four time-of-day components for traffic. The declared partition  $\mathcal{D}$  has one block per source group. A transported initial-



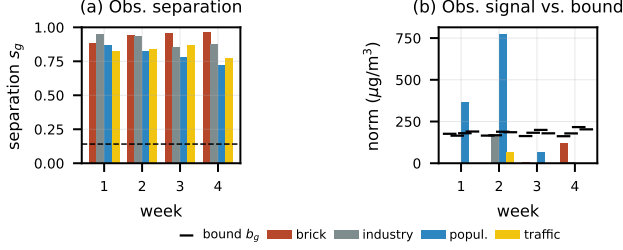


Figure 3: Observed New Delhi, weeks 1–4. **(a)** Separation  $s_g$  of each source group, with the dashed line at  $\tau_\theta = 0.141$ . **(b)** Norm of each fitted group signal over the week’s readings (bars) and its error bound  $b_g$  of Theorem 5.9(c) at  $\sigma_e = 30.68$ ,  $\delta = 0.05$  (ticks).

condition field is subtracted (Appendix D.6), and there is no ground truth.

**Noise level.** The two co-located monitors at Pusa differ with standard deviation  $43.4 \mu\text{g}/\text{m}^3$  over 17,949 shared hours (none in May 2018). If their errors are independent with equal variance, each has  $\sigma_e = 43.4/\sqrt{2} = 30.7 \mu\text{g}/\text{m}^3$ . The difference has lag-1 autocorrelation 0.857 (Appendix F.9).  $\sigma_e$  describes instrument error at one site, and operator error adds to it.

**Identifiable resolution.** Every week has numerical rank 7 of 7, so  $\mathcal{P}^*$  is the seven single columns and  $\mathcal{P}^* \vee \mathcal{D}$  is the four source groups. No fingerprint is zero,  $\sigma_J$  is 3.71–10.15, and the largest pairwise coherence is 0.316–0.496, so no merge, weak-column,  $\sigma_J$  or coherence flag is raised (absorption was not recorded). Every group has separation  $0.72\text{--}0.96 \geq \tau_\theta$ , so IASA reports the four groups in every week (Figure 3(a)). At  $\sigma_e = 30.68$ ,  $\tau_\theta$  corresponds to  $\eta \approx 1,100 \mu\text{g}/\text{m}^3$ , above every fitted group norm, so the noise statement comes from  $b_g$ , not from the grouping.

**Group signals against the noise level.** Theorem 5.9(c) with  $\sigma_e = 30.68$ ,  $\text{rank}(\tilde{A}) = 7$  and  $\delta = 0.05$  bounds the error of each group’s fitted signal by  $b_g = 156.2/s_g$ , which is  $162\text{--}217 \mu\text{g}/\text{m}^3$  for the four groups (Figure 3(b)). For the largest group,  $b_g$  is 0.49 and 0.24 times its fitted norm in weeks 1 and 2 (population, with norms 365 and 776) and 2.96 and 1.33 times it in weeks 3 and 4 (population with norm 67.5 and brick kilns with norm 121.5). At this noise level the bound therefore establishes that the largest group’s signal is resolved in weeks 1 and 2, and does not establish it in weeks 3 and 4. The bootstrap gives the same split (Table 17): the 95% interval of the largest share has width 0.25 and 0.14 in weeks 1 and 2, and 0.94 and 0.67 in weeks 3 and 4. For AR(1) errors with coefficient 0.857, independent across sensors,  $\|\Sigma\|_2 \leq (1 + 0.857)/(1 - 0.857) \sigma_e^2 = 13.0 \sigma_e^2$ , a level of

110.7, at which the bound exceeds the largest group’s norm in every week except week 2.

**What the fit explains.** The projected residual has root mean square  $46\text{--}91 \mu\text{g}/\text{m}^3$  per reading, against  $1.05\text{--}13.24$  for the total fitted signal ( $2.3\text{--}14.5\%$  of the residual in each week). The adequacy test reports no verdict because no calibrated noise model exists. The bounds above hold under Assumptions 5.1 and 5.2, and a residual root mean square  $1.5\text{--}3.0$  times  $\sigma_e$  means that the declared model, the noise level, or both do not describe these weeks fully. The shares are fractions of the fitted projected concentration signal, so they are not emission shares, and because the nuisance terms are removed first they are not shares of  $\text{PM}_{2.5}$  mass.

## 7 CONCLUSION

We attributed  $\text{PM}_{2.5}$  from a regulatory network to declared source groups at the resolution that the projected transport operator supports. The finest exactly identifiable grouping is unique and is computed before fitting, and the error of each reported group’s signal is bounded by the noise divided by its separation. In controlled experiments the procedure flagged the two situations in which the mass balance without background removal had group-signal errors of 36–784%. On the observed New Delhi weeks, the four source groups are separable in every week, but at the instrument noise level the error bound of the largest group’s signal is below its fitted norm in two of four weeks and above it in the other two. With historical wind, the diagnostics can assess a sensor layout before deployment.

**Limitations.** All results are conditional on the declared wind, transport model, inventories and nuisance basis. The shares are fractions of the fitted projected concentration signal, not shares of emissions or of  $\text{PM}_{2.5}$  mass, and on the observed weeks the fitted signal is  $2.3\text{--}14.5\%$  of the residual in root mean square. An omitted source can be absorbed by the fit without detection. The noise level comes from one pair of co-located monitors over a record that excludes the study weeks, and the bounds and the bootstrap assume Gaussian noise and an exact operator. The transport model is two-dimensional, without mixing height, deposition or secondary formation, and dust, biomass burning and secondary aerosol are not declared. Sensitivity of the observed results to the lag, nuisance basis and dispersion is not reported, the gridded wind and the inventory crop’s alignment are not verified, and the controlled experiments do not compare  $b_g$  or bootstrap coverage with known errors. The observed study covers four weeks of May 2018 without ground truth. The network’s gas measurements are not used and could serve as additional tracers.



## AI use statement

In this work, we used generative AI tools (Claude, Anthropic) for writing code (the grouping module and its tests), for restructuring and editing the manuscript, for literature search and checking bibliographic entries, and for drafting proofs and checking them numerically on random instances. We have not used generative AI tools to generate, modify or label data. All AI-assisted code was run and tested, every reported number was regenerated from the released code and the committed results, and the proofs were checked by the authors. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

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## CHECKLIST

1. For all models and algorithms presented, check if you include:
  - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes; Sections 3 and 5, Algorithm 1, Appendix C.]
  - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes; Theorem 5.7 and the text after it ( $O(NJ^2)$  for the exact components).]
  - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes; supplementary archive with `model/iasa`, `experiments/` and the dependency list (Python 3.12, PyTorch, NumPy, SciPy, pandas, scikit-learn, einops, Matplotlib).]
2. For any theoretical claim, check if you include:
  - (a) Statements of the full set of assumptions of all theoretical results. [Yes; each statement in Section 5 and Appendix A lists its assumptions.]
  - (b) Complete proofs of all theoretical results. [Yes; Appendix A.]
  - (c) Clear explanations of any assumptions. [Yes.]
3. For all figures and tables that present empirical results, check if you include:
  - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes; supplementary code and configurations; the  $PM_{2.5}$  and wind records are public CPCB data (CPCB, 2023, 2025).]
  - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes; thresholds in Table 3, solver settings in Appendix K.]
  - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes; metrics are defined where reported; Experiment 11 is over three seeds, and the observed shares carry 95% parametric-bootstrap intervals.]
  - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes; Appendix M: Experiment 11 and the observed re-analysis ran on one laptop CPU (Apple M5); the other controlled experiments and the observed fits ran on an NVIDIA GPU with CUDA 11.8, and the controlled experiments reproduce on CPU.]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
  - (a) Citations of the creator If your work uses existing assets. [Yes; CPCB data, emission inventories and the population grid are cited.]
  - (b) The license information of the assets, if applicable. [Yes; Appendix M.]
  - (c) New assets either in the supplemental material or as a URL, if applicable. [Yes; code in the supplementary archive.]
  - (d) Information about consent from data providers/curators. [Not Applicable; public government records.]
  - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable.]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
  - (a) The full text of instructions given to participants and screenshots. [Not Applicable.]
  - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable.]
  - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable.]

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# Identifiability-Aware Source Attribution of Air Pollution: Supplementary Materials

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## A PROOFS

In this appendix  $A$  denotes the projected operator  $\tilde{A}$  of Equation 3,  $\mathbf{a}_j$  its columns,  $V_g = \text{col}(A_g)$ ,  $W_g = \text{col}(A_{E \setminus g})$  and  $U = \text{col}(A)$ . The unit tests in the code release (`tests/test_iasa_grouping.py`) check numerically the exact components against brute force (Theorem 5.7(b)–(d)), the two-column separation  $\sqrt{1 - \rho^2}$ , Theorem 5.9(a), the examples of Proposition A.2, and the exhaustive and greedy partition searches.

### A.1 Individual Contributions

**Proof of Proposition 5.3.** If  $\text{rank}(A) = J$ , the rank-nullity theorem gives  $\ker(A) = \{\mathbf{0}\}$ , so  $A\mathbf{c}_1 = A\mathbf{c}_2$  implies  $\mathbf{c}_1 = \mathbf{c}_2$ . Conversely, if  $\text{rank}(A) < J$  there is a nonzero  $\mathbf{d} \in \ker(A)$ . Choose  $\beta > \|\mathbf{d}\|_\infty$  and set  $\mathbf{c} = \beta\mathbf{1}$ . Then  $\mathbf{c}$  and  $\mathbf{c} + \mathbf{d}$  are distinct, both lie in  $\mathbb{R}_+^J$  because  $\beta + d_j \geq \beta - \|\mathbf{d}\|_\infty > 0$ , and  $A(\mathbf{c} + \mathbf{d}) = A\mathbf{c}$ .  $\square$

The same interior-point construction is used below whenever a kernel vector must be turned into two nonnegative vectors with equal projected observations.

**Proof of Proposition 5.4.** For least squares with full column rank,  $\hat{\mathbf{c}} - \mathbf{c} = A^\dagger \tilde{\boldsymbol{\epsilon}} = A^\dagger P_A \tilde{\boldsymbol{\epsilon}}$ , and  $\|A^\dagger\|_2 = 1/\sigma_J(A)$ . For NNLS, let  $C = \{A\mathbf{z} : \mathbf{z} \geq 0\}$ , a closed convex cone contained in  $U$ . For  $\mathbf{x} \in U$ ,  $\|\tilde{\mathbf{y}} - \mathbf{x}\|_2^2 = \|P_A \tilde{\mathbf{y}} - \mathbf{x}\|_2^2 + \|(I - P_A)\tilde{\mathbf{y}}\|_2^2$ , so the fitted signal is  $A\hat{\mathbf{c}} = \Pi_C(\tilde{\mathbf{y}}) = \Pi_C(P_A \tilde{\mathbf{y}})$ , where  $\Pi_C$  is the Euclidean projection onto  $C$ . Since  $\mathbf{c} \geq 0$ ,  $A\mathbf{c} \in C$  and  $\Pi_C(A\mathbf{c}) = A\mathbf{c}$ . Projection onto a closed convex set is nonexpansive, so

$$\|A(\hat{\mathbf{c}} - \mathbf{c})\|_2 = \|\Pi_C(P_A \tilde{\mathbf{y}}) - \Pi_C(A\mathbf{c})\|_2 \leq \|P_A \tilde{\mathbf{y}} - A\mathbf{c}\|_2 = \|P_A \tilde{\boldsymbol{\epsilon}}\|_2.$$

Full column rank gives  $\|A\Delta\|_2 \geq \sigma_J(A)\|\Delta\|_2$  for  $\Delta = \hat{\mathbf{c}} - \mathbf{c}$ , which proves the first inequality in (6); the second holds because  $P_A$  is an orthogonal projector.  $\square$

**Ray distance (Equation 9).** For nonzero  $\mathbf{a}_i, \mathbf{a}_j$ , the function  $\alpha \mapsto \|\mathbf{a}_i - \alpha \mathbf{a}_j\|_2^2 / \|\mathbf{a}_i\|_2^2$  is strictly convex with minimizer  $\alpha^* = \mathbf{a}_i^\top \mathbf{a}_j / \|\mathbf{a}_j\|_2^2$  and minimum  $1 - \rho_{ij}^2$ . The minimum is also  $\sin^2$  of the angle between  $\mathbf{a}_i$  and  $\text{span}(\mathbf{a}_j)$ , so for  $J = 2$  the ray distance equals the sine of the angle between the two columns.  $\square$

**Proof of Proposition D.1.** Let  $\hat{A}$  have full column rank and  $\|\hat{A} - A^*\|_2 \leq \delta_H$ . The data obey  $\tilde{\mathbf{y}} = A^* \mathbf{c} + \tilde{\boldsymbol{\epsilon}} = \hat{A} \mathbf{c} + [\tilde{\boldsymbol{\epsilon}} + (A^* - \hat{A})\mathbf{c}]$ , and  $\hat{A}^\dagger \hat{A} = I_J$ , so  $\hat{\mathbf{c}} - \mathbf{c} = \hat{A}^\dagger [\tilde{\boldsymbol{\epsilon}} + (A^* - \hat{A})\mathbf{c}]$ . Taking norms gives Equation 26.  $\square$

### A.2 Identifiable Groupings

**Proof of Theorem 5.7.** (a) Suppose  $U = \bigoplus_g V_g$ . If  $A\mathbf{c}_1 = A\mathbf{c}_2$ , then  $\sum_g A_g(\mathbf{c}_{1,g} - \mathbf{c}_{2,g}) = 0$  with each term in  $V_g$ , so each term is zero. Conversely, if the sum  $\sum_g V_g$  is not direct, there are  $\mathbf{u}_g \in V_g$ , not all zero, with  $\sum_g \mathbf{u}_g = 0$ . Writing  $\mathbf{u}_g = A_g \mathbf{d}_g$  defines  $\mathbf{d} \in \ker(A)$  with  $A_h \mathbf{d}_h \neq 0$  for some  $h$ . With  $\mathbf{c} = \beta\mathbf{1}$  and  $\beta > \|\mathbf{d}\|_\infty$ , the vectors  $\mathbf{c}$  and  $\mathbf{c} + \mathbf{d}$  are nonnegative, have equal projected observations, and have different signals in block  $h$ . The rank form holds because  $\dim \sum_g V_g = \text{rank}(A)$ , with equality to  $\sum_g \dim V_g$  exactly when the sum is direct.

(b, c) Let  $B$  index a basis of  $U$  chosen from the columns, write  $\mathbf{a}_e = \sum_{b \in B} T_{be} \mathbf{a}_b$  for  $e \notin B$ , and let  $S_1, \dots, S_m$  be the vertex sets of the connected components of the graph that joins  $e \notin B$  to every  $b$  with  $T_{be} \neq 0$  (isolated vertices are components).

*Step 1:*  $\mathcal{S} = \{S_i\}$  is identifiable. Let  $B_i = B \cap S_i$ . Every  $e \in S_i \setminus B$  has all basis columns with nonzero coefficient in  $S_i$ , so  $\mathbf{a}_e \in \text{span}\{\mathbf{a}_b : b \in B_i\}$  and  $\text{rank}(A_{S_i}) = |B_i|$ . Hence  $\sum_i \text{rank}(A_{S_i}) = |B| = \text{rank}(A)$ , and (a) applies.

*Step 2:* every identifiable  $\mathcal{P}$  coarsens  $\mathcal{S}$ . Suppose some  $S_i$  meets two blocks of  $\mathcal{P}$ . Since  $S_i$  is connected, some edge  $\{e, b\}$  with  $T_{be} \neq 0$  has  $e \in g$  and  $b \in h$  for distinct blocks  $g, h$ . Let  $d_e = 1$ ,  $d_{b'} = -T_{b'e}$  for  $b' \in B$ , and

$d_j = 0$  otherwise; then  $A\mathbf{d} = 0$ , and  $A_h\mathbf{d}_h = -\sum_{b' \in B \cap h} T_{b'e} \mathbf{a}_{b'}$  is a combination of linearly independent columns with the nonzero coefficient  $-T_{be}$ , so  $A_h\mathbf{d}_h \neq 0$ . The interior-point construction contradicts identifiability.

*Step 3: coarsening preserves identifiability.* Grouping the summands of a direct sum gives a direct sum, and  $V_{g \cup h} = V_g + V_h$ .

Steps 1–3 show that the identifiable partitions are exactly the coarsenings of  $\mathcal{S}$ , so  $\mathcal{S}$  is the unique finest identifiable partition and does not depend on  $B$ . This proves (b) and the graph characterization in (c). That the components of this graph are the connected components of the column matroid is Krogdahl (1977); see also Oxley (2011).

(d) A partition whose blocks are unions of blocks of  $\mathcal{D}$  is identifiable exactly when it is a common coarsening of  $\mathcal{P}^*$  and  $\mathcal{D}$ . The finest common coarsening  $\mathcal{P}^* \vee \mathcal{D}$  is one, by Step 3.  $\square$

**Proposition A.1** (Sums of coefficients). *Call  $\mathcal{P}$  sum-identifiable if  $A\mathbf{c}_1 = A\mathbf{c}_2$  with  $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}_+^J$  implies  $\sum_{j \in g} c_{1,j} = \sum_{j \in g} c_{2,j}$  for every block  $g$ . This holds if and only if every block indicator lies in  $\text{row}(A)$ ; minimal sum-identifiable partitions need not be unique, and for rational  $A$  deciding whether one with at least two blocks exists is NP-complete.*

**Proof of Proposition A.1.**  $\mathcal{P}$  is sum-identifiable when  $\mathbf{1}_g^\top \mathbf{c}$  is determined by  $A\mathbf{c}$  for every block, on  $\mathbb{R}_+^J$ . By the interior-point construction, a linear functional  $\boldsymbol{\lambda}^\top \mathbf{c}$  is determined exactly when  $\boldsymbol{\lambda} \perp \ker(A)$ , that is  $\boldsymbol{\lambda} \in \text{row}(A)$ , so  $\mathcal{P}$  is sum-identifiable exactly when every block indicator lies in  $\text{row}(A)$ .

*Non-uniqueness.* Let  $A$  have rows  $(1, 1, 0, 0)$ ,  $(0, 0, 1, 1)$  and  $(1, 0, 1, 0)$ . Its row space contains  $(0, 1, 0, 1)$  but no unit vector and not  $(1, 0, 0, 1)$ , so the sum-identifiable partitions are  $\{E\}$ ,  $\{\{1, 2\}, \{3, 4\}\}$  and  $\{\{1, 3\}, \{2, 4\}\}$ ; the last two are both minimal. For the signal target on the same  $A$ ,  $\ker(A)$  is spanned by  $(1, -1, -1, 1)$ , the four columns form one circuit, and  $\mathcal{P}^* = \{E\}$ .

*NP-completeness.* The problem “given rational  $A$ , is there a sum-identifiable partition with at least two blocks?” is in NP, since a partition can be checked with rank computations. For hardness, take an instance  $w_1, \dots, w_n$  of PARTITION (Garey and Johnson, 1979) with  $\sum_i w_i = 2T$ , let  $\mathbf{d} = (w_1, \dots, w_n, -T, -T) \in \mathbb{Z}^{n+2}$ , and let  $A$  be a rational  $(n+1) \times (n+2)$  matrix whose rows span  $\mathbf{d}^\perp$ , so that  $\text{row}(A) = \mathbf{d}^\perp$ . Because  $\mathbf{1}_E^\top \mathbf{d} = 0$ , a partition with at least two blocks exists exactly when some nonempty proper  $g \subset E$  has  $\mathbf{1}_g^\top \mathbf{d} = 0$  (then  $\{g, E \setminus g\}$  qualifies). If  $g$  contains neither of the last two coordinates,  $\mathbf{1}_g^\top \mathbf{d} > 0$ ; if it contains both,  $\mathbf{1}_g^\top \mathbf{d} = 0$  forces  $g = E$ ; if it contains exactly one, the condition is  $\sum_{i \in g, i \leq n} w_i = T$ , a solution of PARTITION.  $\square$

### A.3 Structure of Reportable Partitions

**Proposition A.2** (Structure of reportable partitions).

- (a) *The one-block partition  $\{E\}$  is reportable at every  $\tau_\theta$ .*
- (b) *If  $g$  and  $h$  are blocks of an identifiable partition, then  $1/s_{g \cup h} \leq 1/s_g + 1/s_h$ , and reportable partitions are not closed under coarsening.*
- (c) *Minimal reportable partitions need not be unique.*

**Remark A.3** (Implication). Proposition A.2 needs only Assumption 5.1. By (a) a reportable partition always exists: in the worst case IASA reports the total signal of all sources. By (b), merging two blocks does not always help: the merged block can be closer to the remaining sources than either part, so IASA checks every union of atoms rather than merging the least separated pair. By (c), more than one partition can have the most blocks; IASA therefore states its tie-break (the largest smallest separation) and lists the remaining ties.

### A.4 An Oblique Projection Lemma

**Lemma A.4.** *Let  $U = V \oplus W$  with  $V \neq \{0\} \neq W$ , let  $\theta \in (0, \pi/2]$  be the smallest principal angle between  $V$  and  $W$ , and let  $\pi$  be the projection onto  $V$  along  $W$ , defined on  $U$ . Then  $\|\pi\|_2 = 1/\sin \theta$ .*

*Proof.* Write  $\mathbf{u} = \mathbf{x} + \mathbf{w}$  with  $\mathbf{x} \in V$ ,  $\mathbf{w} \in W$ . By the definition of  $\theta$ ,  $|\mathbf{x}^\top \mathbf{w}| \leq \cos \theta \|\mathbf{x}\|_2 \|\mathbf{w}\|_2$ , so

$$\|\mathbf{u}\|_2^2 \geq \|\mathbf{x}\|_2^2 + \|\mathbf{w}\|_2^2 - 2 \cos \theta \|\mathbf{x}\|_2 \|\mathbf{w}\|_2 = (\|\mathbf{w}\|_2 - \cos \theta \|\mathbf{x}\|_2)^2 + \sin^2 \theta \|\mathbf{x}\|_2^2 \geq \sin^2 \theta \|\mathbf{x}\|_2^2,$$

which gives  $\|\pi \mathbf{u}\|_2 \leq \|\mathbf{u}\|_2 / \sin \theta$ . For unit principal vectors  $\mathbf{x}_0 \in V$ ,  $\mathbf{w}_0 \in W$  with  $\mathbf{x}_0^\top \mathbf{w}_0 = \cos \theta$ , the vector  $\mathbf{u} = \mathbf{x}_0 - \cos \theta \mathbf{w}_0$  has  $\|\mathbf{u}\|_2 = \sin \theta$  and  $\pi \mathbf{u} = \mathbf{x}_0$ , so the bound is attained (Ipsen and Meyer, 1995).  $\square$

If  $V = \{0\}$ , then  $\pi = 0$ ; if  $W = \{0\}$ ,  $\pi$  is the identity on  $U$ . Both cases are consistent with the convention  $s_g = 1$ .

### A.5 Group-Signal Error, Reportable Partitions and Stability

**Proof of Theorem 5.9.** (a) By Theorem 5.7(a),  $U = \bigoplus_g V_g$ , and  $W_g = \bigoplus_{h \neq g} V_h$ . Let  $\pi_g$  be the projection of  $U$  onto  $V_g$  along  $W_g$ . For any  $\mathbf{z}$ ,  $A\mathbf{z} = \sum_g A_g \mathbf{z}_g$  with  $A_g \mathbf{z}_g \in V_g$ , so  $A_g \mathbf{z}_g = \pi_g(A\mathbf{z})$ . All least-squares solutions have the fitted signal  $P_A \tilde{\mathbf{y}}$ , and all NNLS solutions have the fitted signal  $\Pi_C(\tilde{\mathbf{y}})$ , which is unique because  $C$  is closed and convex; hence the group signals do not depend on the solution. Moreover  $A_g(\hat{\mathbf{c}}_g - \mathbf{c}_g) = \pi_g(A(\hat{\mathbf{c}} - \mathbf{c}))$ , where  $A(\hat{\mathbf{c}} - \mathbf{c}) = P_A \tilde{\mathbf{e}}$  for least squares and  $\|A(\hat{\mathbf{c}} - \mathbf{c})\|_2 \leq \|P_A \tilde{\mathbf{e}}\|_2$  for NNLS (proof of Proposition 5.4). Lemma A.4 with  $V = V_g$ ,  $W = W_g$  gives  $\|\pi_g\|_2 \leq 1/s_g$ , with equality when  $V_g$  and  $W_g$  are nonzero, which proves (7).

(b) Since  $U \subseteq \text{col}(Q)^\perp$ ,  $P_A \tilde{\mathbf{e}} = P_A P_Q^\perp \boldsymbol{\epsilon} = P_A \boldsymbol{\epsilon}$ . The least-squares error of block  $g$  is  $M\boldsymbol{\epsilon}$  with  $M = \pi_g P_A$ , a matrix of rank  $r_g$  with  $\|M\|_2 = \|\pi_g\|_2 = 1/s_g$  when  $r_g \geq 1$ ; when  $r_g = 0$ ,  $M = 0$  and the error is zero. Hence  $\mathbb{E}\|M\boldsymbol{\epsilon}\|_2^2 = \sigma^2 \|M\|_F^2 \leq \sigma^2 r_g \|M\|_2^2 = \sigma^2 r_g / s_g^2$ , with equality when  $r_g = 1$  because  $M$  then has one nonzero singular value. The map  $\boldsymbol{\epsilon} \mapsto \|M\boldsymbol{\epsilon}\|_2$  is  $\|M\|_2$ -Lipschitz, so the Gaussian concentration inequality of Tsirelson, Ibragimov and Sudakov (Boucheron et al., 2013) gives  $\|M\boldsymbol{\epsilon}\|_2 \leq \mathbb{E}\|M\boldsymbol{\epsilon}\|_2 + \sigma \|M\|_2 \sqrt{2 \log(1/\delta)}$  with probability at least  $1 - \delta$ , and  $\mathbb{E}\|M\boldsymbol{\epsilon}\|_2 \leq \sigma \|M\|_F \leq \sigma \sqrt{r_g / s_g}$ ; this gives a high-probability least-squares bound with  $\sqrt{r_g}$  in place of  $\sqrt{\text{rank}(A)}$ , which the theorem does not state.

(c) As in (b),  $P_A \tilde{\mathbf{e}} = P_A \boldsymbol{\epsilon}$ . Write  $\boldsymbol{\epsilon} = \Sigma^{1/2} \mathbf{z}$  with  $\mathbf{z} \sim N(0, I_N)$  ( $\Sigma = \sigma^2 I$  under Assumption 5.2) and  $f(\mathbf{z}) = \|P_A \Sigma^{1/2} \mathbf{z}\|_2$ . The map  $f$  is Lipschitz with constant  $\|P_A \Sigma^{1/2}\|_2 \leq \|\Sigma\|_2^{1/2}$ , and  $\mathbb{E}f \leq (\mathbb{E}f^2)^{1/2} = \|P_A \Sigma^{1/2}\|_F \leq \sqrt{\text{rank}(A)} \|\Sigma\|_2^{1/2}$ , because  $P_A \Sigma^{1/2}$  has rank at most  $\text{rank}(A)$ . The Gaussian concentration inequality (Boucheron et al., 2013) gives, with probability at least  $1 - \delta$ ,  $\|P_A \tilde{\mathbf{e}}\|_2 \leq \|\Sigma\|_2^{1/2} (\sqrt{\text{rank}(A)} + \sqrt{2 \log(1/\delta)})$ . On this event, part (a), which holds for every noise vector, applies to every block of every identifiable partition at once.  $\square$

**Proof of Proposition A.2.** (a)  $\{E\}$  is identifiable and  $s_E = 1$  by convention. (b) With  $\pi_g$  as above,  $\pi_{g \cup h} = \pi_g + \pi_h$ , because both send  $\sum_k \mathbf{x}_k$  (with  $\mathbf{x}_k \in V_k$ ) to  $\mathbf{x}_g + \mathbf{x}_h$ . By Lemma A.4,  $1/s_{g \cup h} = \|\pi_g + \pi_h\|_2 \leq 1/s_g + 1/s_h$  when  $V_{g \cup h} \neq \{0\}$ ; when  $V_{g \cup h} = \{0\}$ ,  $s_{g \cup h} = 1$  and the inequality holds because  $1/s_g, 1/s_h \geq 1$ . For the  $4 \times 4$  operator

$$\begin{pmatrix} 0.248 & -0.142 & 0.165 & 0.828 \\ 0.899 & -0.775 & -0.917 & 0.229 \\ -0.356 & -0.381 & 0.243 & -0.507 \\ -0.063 & -0.484 & -0.270 & 0.069 \end{pmatrix}$$

the four singletons have separations of at least 0.3, while the blocks of  $\{\{1, 2\}, \{3, 4\}\}$  have separation below 0.3, so at  $\tau_\theta = 0.3$  a reportable partition has a coarsening that is not reportable. (c) Let  $\tilde{\mathbf{a}}_1 = (1, 0, 0)$ ,  $\tilde{\mathbf{a}}_2 = (\cos a, \sin a, 0)$  and  $\tilde{\mathbf{a}}_3 = (\cos a, 0, \sin a)$  with  $a = 0.3$ , and  $\tau_\theta = 0.25$ .  $\text{span}(\tilde{\mathbf{a}}_1, \tilde{\mathbf{a}}_2)$  is the plane  $x_3 = 0$ , at angle  $a$  from  $\tilde{\mathbf{a}}_3$ , so both blocks of  $\{\{1, 2\}, \{3\}\}$  have separation  $\sin a = 0.296$ ; by symmetry so do those of  $\{\{1, 3\}, \{2\}\}$ . The angle between  $\tilde{\mathbf{a}}_1$  and  $\text{span}(\tilde{\mathbf{a}}_2, \tilde{\mathbf{a}}_3)$  has sine  $\sin a / \sqrt{1 + \cos^2 a} = 0.214$ , which is  $s_1$  and also the separation of  $\{2, 3\}$ . At  $\tau_\theta = 0.25$  the partitions  $\{\{1, 2\}, \{3\}\}$  and  $\{\{1, 3\}, \{2\}\}$  are reportable, neither refines the other, and the only finer partition, the three singletons, is not reportable because  $s_1 = 0.214$ .  $\square$

**Minimal reportable partitions.** Let the atoms partition  $E$ . A partition into unions of atoms whose blocks all have  $s_h > 0$  is identifiable:  $s_h > 0$  means  $V_h \cap W_h = \{0\}$  with  $W_h = \sum_{k \neq h} V_k$ , and a sum of subspaces is direct exactly when each summand meets the sum of the others only in  $\{0\}$ . So a partition into unions of atoms is reportable exactly when every block has  $s_h \geq \tau_\theta$ . Since  $s_h$  depends only on  $h$ , a reportable partition has a strictly finer reportable partition into unions of atoms exactly when one of its blocks can be split into unions of atoms that all have separation at least  $\tau_\theta$ . This gives the characterization used in step 5 of Algorithm 1.

**Computing minimal reportable partitions.** With  $q$  atoms, the separations of all nonempty proper unions of atoms take  $2^{q-1} - 1$  principal-angle computations, because  $s_h = s_{E \setminus h}$  and  $s_E = 1$ . From the flags  $s_h \geq \tau_\theta$ , dynamic programming over subsets that contain their lowest atom decides for every union  $S$  whether it can be partitioned into reportable unions, and hence which reportable unions cannot be split, in  $O(3^q)$  time. The same recursion, maximizing first the number of blocks and then the smallest separation, returns the reported partition in  $O(3^q)$  time, because  $\min(s_h, \cdot)$  preserves the order of the second criterion. Listing all minimal reportable partitions, which IASA does to report ties, adds work for each listed partition. The greedy rule merges the least separated block with the partner that maximizes the separation of the merged block; merging  $g$  and  $h$  leaves every other block's separation unchanged, because  $s_k$  depends only on  $k$ , and the rule stops within  $q - 1$  merges

Table 2: Identifiability diagnostics and reporting outputs produced with source activity estimates. The lower block lists the set- and graph-valued outputs that accompany the scalar diagnostics.

| Symbol                      | Name                       | Interpretation                  | Action                           |
|-----------------------------|----------------------------|---------------------------------|----------------------------------|
| $r_{\text{num}}(\tilde{A})$ | numerical rank             | exact separability              | require rank $J$                 |
| $\sigma_J(\tilde{A})$       | padded min. singular value | noise robustness                | flag zero/small                  |
| $\kappa(\tilde{A})$         | condition number           | error amplification             | $\infty$ if deficient            |
| $r_{\text{eff}}$            | effective rank             | noise-level resolution          | reduce grouping                  |
| $v_j$                       | coefficient visibility     | measurable effect               | flag weak                        |
| $a_j$                       | background absorption      | signal lost to $Q$              | warn after projection            |
| $\rho_{ij}$                 | pairwise coherence         | pairwise similarity             | merge if $> \tau_\rho$           |
| $d_{ij}^{\text{ray}}$       | ray distance               | $\sqrt{1 - \rho_{ij}^2}$        | N/A if weak                      |
| $s_g$                       | separation                 | noise-level separability of $g$ | report $g$ if $\geq \tau_\theta$ |
| $\mathcal{W}$               | weak set                   | undetectable coefficients       | flag                             |
| $\mathcal{P}^*$             | matroid components         | exact resolution                | atoms with $\mathcal{D}$         |

because  $s_E = 1$ . It has no optimality guarantee.

**Proof of Proposition 5.12.** For unit vectors let  $d(\mathbf{x}, \mathbf{y}) = \arccos |\mathbf{x}^\top \mathbf{y}|$ , the angle between the lines they span, which is a metric on lines. Then  $\theta_g = \min\{d(\mathbf{x}, \mathbf{w}) : \mathbf{x} \in V_g, \mathbf{w} \in W_g\}$ . For a full-column-rank  $M$  and  $\|\Delta\|_2 \leq \sigma_{\min}(M)/2$ , so that the arcsine arguments below are at most 1: a unit  $\mathbf{x}' = (M + \Delta)\mathbf{z}/\|(M + \Delta)\mathbf{z}\|_2$  in  $\text{col}(M + \Delta)$  is within distance  $\|\Delta\mathbf{z}\|_2/\|(M + \Delta)\mathbf{z}\|_2 \leq \|\Delta\|_2/(\sigma_{\min}(M) - \|\Delta\|_2)$  of  $\text{col}(M)$ , and a unit  $\mathbf{x} = M\mathbf{z}/\|M\mathbf{z}\|_2$  is within distance  $\|\Delta\|_2/\sigma_{\min}(M)$  of  $\text{col}(M + \Delta)$ . For a unit vector the distance to a subspace is the sine of its angle to that subspace, so in both directions the nearest line is within angle  $\arcsin(\|\Delta\|_2/(\sigma_{\min}(M) - \|\Delta\|_2))$ . Applied to  $M = A_g$  and  $M = A_{E \setminus g}$  this gives the angles  $\beta_g$  and  $\beta_{E \setminus g}$ . If  $\mathbf{x}', \mathbf{w}'$  attain  $\theta_g(A')$  and  $\mathbf{x} \in V_g, \mathbf{w} \in W_g$  are their nearest lines, the triangle inequality gives  $\theta_g(A) \leq d(\mathbf{x}, \mathbf{w}) \leq \beta_g + \beta_{E \setminus g} + \theta_g(A')$ ; the same argument from  $A$  to  $A'$  gives the reverse inequality. For the second statement,  $\|\Delta\|_2 \leq \sigma_J(A)/2$  gives  $\sigma_J(A') \geq \sigma_J(A)/2 > 0$ , so both operators have full column rank, every partition is identifiable for both, and both have  $\mathcal{P}^*$  equal to the singletons, so the atoms coincide. For a nonempty proper union  $g$  of atoms,  $\sigma_{\min}(A_g) \geq \sigma_J(A) \geq 2\|\Delta\|_2 \geq 2\|\Delta_g\|_2$ , and the same for  $E \setminus g$ , so the first statement applies and the margin condition keeps  $\theta_g$  on the same side of  $\arcsin \tau_\theta$ ;  $s_E = 1$  for both. By the characterization of minimal reportable partitions above, the reportable and the minimal reportable partitions coincide.  $\square$

## B IDENTIFIABILITY DIAGNOSTICS: DEFINITIONS

This appendix gives the full definitions of the diagnostics summarized in Section 5 and Table 2. It uses the general notation of Section 3; “background” refers to the nuisance basis  $Q$ , and a “coefficient” is one component  $j$ .

For floating-point computation, the default numerical rank tolerance is  $\tau_{\text{num}} = \max(N, J) \epsilon_{\text{mach}} \sigma_1$  with numerical rank  $r_{\text{num}} = \#\{i : \sigma_i > \tau_{\text{num}}\}$ , where  $\epsilon_{\text{mach}}$  is the machine precision of the SVD dtype (about  $2.2 \times 10^{-16}$  in double precision). This  $\tau_{\text{num}}$  is a purely numerical tolerance for detecting exact rank deficiency; it is distinct from the scientific, noise-dependent threshold  $\tau_\sigma$  below. The primary identifiability score is the *padded minimum singular value*  $\sigma_J(\tilde{A})$ , where the singular spectrum is padded with zeros whenever  $J > N$  or  $\tilde{A}$  is rank deficient. A rank-deficient response therefore scores zero and is never reported as stable. The condition number is  $\kappa(\tilde{A}) = \sigma_1/\sigma_J$  when  $r_{\text{num}} = J$  and  $\infty$  when  $r_{\text{num}} < J$ . The smallest numerically positive value may additionally be reported as  $\sigma_{\min,+}$ , but it is not the identifiability score and never turns a rank-deficient matrix into a stable one. For a separately chosen, noise-dependent threshold  $\tau_\sigma$ , the effective rank is  $r_{\text{eff}}(\tau_\sigma) = \#\{i : \sigma_i(\tilde{A}) > \tau_\sigma\}$ . Unlike  $\tau_{\text{num}}$ , the threshold  $\tau_\sigma$  is meant to be set from the observation-noise scale, not from machine precision; it is declared before fitting and is not tuned on source-recovery results. When no  $\tau_\sigma$  is declared, the implementation sets  $\tau_\sigma = \tau_{\text{num}}$ , so that  $r_{\text{eff}}$  equals the numerical rank; this is the case in every reported run.

For coefficient-specific diagnostics, let  $\tilde{\mathbf{a}}_j, j = (k, b)$ , be a projected source–basis fingerprint. Its visibility and the weak set at threshold  $\tau_v \geq 0$  are  $v_j = \|\tilde{\mathbf{a}}_j\|_2$  and  $\mathcal{W} = \{j : v_j \leq \tau_v\}$ . A weak coefficient has little measurable effect after transport and background correction:  $v_j$  is the sensor-time magnitude produced by one unit of coefficient  $c_j$ , and  $\tau_v$  marks fingerprints below the minimum detectable signal. A practical predeclared rule sets  $\tau_v$  from the noise scale and a target signal-to-noise ratio, e.g.  $\tau_v = \text{SNR}_{\min} \sigma_e$ , so a unit coefficient is flagged weak



Table 3: IASA thresholds, their meaning, and how they are set. None is optimized on source-recovery error.

| Threshold           | Meaning              | How set                                                                                                                                  |
|---------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $\tau_{\text{num}}$ | numerical rank tol.  | $\max(N, J)\epsilon_{\text{mach}}\sigma_1$                                                                                               |
| $\tau_{\sigma}$     | effective-rank floor | declared noise scale; $\tau_{\text{num}}$ if none                                                                                        |
| $\tau_v$            | visibility floor     | e.g. $\text{SNR}_{\min}\sigma_e$ ; 0 in every reported run                                                                               |
| $\tau_{\theta}$     | separation threshold | $\sqrt{1 - 0.99^2} = 0.141$ in every reported run, or<br>$\sigma(\sqrt{\text{rank } \tilde{A}} + \sqrt{2\log(1/\delta)})/\eta$           |
| $\tau_{\rho}$       | pairwise coherence   | 0.99; for two columns $\rho > \tau_{\rho}$ exactly when $s < \tau_{\theta}$                                                              |
| flag                | identifiability flag | merge, nonempty $\mathcal{W}$ , visibility ratio $\leq 0.1$ , absorption $\geq 0.9$ , $\sigma_J \leq 10^{-6}$ , or coherence $\geq 0.99$ |
| $\epsilon_w$        | wind-drift cap       | physical wind tolerance                                                                                                                  |

when even its full response is indistinguishable from noise. Every reported run uses  $\tau_v = 0$ , so only an exactly zero fingerprint is weak. For eligible indices  $i, j \notin \mathcal{W}$ , pairwise ambiguity is measured by projected fingerprint coherence  $\rho_{ij} = |\tilde{\mathbf{a}}_i^{\top} \tilde{\mathbf{a}}_j| / (\|\tilde{\mathbf{a}}_i\|_2 \|\tilde{\mathbf{a}}_j\|_2)$ . Values near one indicate nearly proportional sensor-time signatures. The ambiguous-pair set is  $\mathcal{A} = \{(i, j) : i, j \notin \mathcal{W}, \rho_{ij} > \tau_{\rho}\}$ ,  $\tau_{\rho} \in [0, 1]$ . The threshold  $\tau_{\rho}$  is predeclared, not optimized after inspecting source recovery. Coherence and ray distance are reported only for eligible (nonweak) pairs; entries involving a weak fingerprint are shown as “N/A” and excluded from extrema, while the weak-visibility flag is always reported.

**Near-dependencies and  $\sigma_J$ .** If a unit vector  $\mathbf{w}$  gives  $\|\sum_j w_j \tilde{\mathbf{a}}_j / \|\tilde{\mathbf{a}}_j\|_2\|_2 = \delta$ , then  $\mathbf{v} = (w_j / \|\tilde{\mathbf{a}}_j\|_2)_j$  has  $\|\tilde{A}\mathbf{v}\|_2 = \delta$  and  $\|\mathbf{v}\|_2 \geq 1 / \max_j \|\tilde{\mathbf{a}}_j\|_2$ , so  $\sigma_J \leq \delta \max_j \|\tilde{\mathbf{a}}_j\|_2$ . A near-dependency among three or more columns with no high pairwise coherence therefore lowers  $\sigma_J$  without triggering the pairwise merge rule.

**Classical connections.** With  $\tilde{\epsilon} = P_Q^{\perp} \epsilon$  and  $\epsilon \sim N(0, \sigma^2 I)$ , the Fisher information for  $\mathbf{c}$  is  $\tilde{A}^{\top} \tilde{A} / \sigma^2$ , because  $\text{col}(\tilde{A}) \perp \text{col}(Q)$ , so the rank condition of Proposition 5.3 is a nonsingular Fisher information (Rothenberg, 1971), and a linear combination  $\boldsymbol{\lambda}^{\top} \mathbf{c}$  is estimable exactly when  $\boldsymbol{\lambda} \in \text{row}(\tilde{A})$  (Searle, 1971). When only components  $i$  and  $j$  are fitted, the correlation between their least-squares estimates is  $-\tilde{\mathbf{a}}_i^{\top} \tilde{\mathbf{a}}_j / (\|\tilde{\mathbf{a}}_i\|_2 \|\tilde{\mathbf{a}}_j\|_2)$ , so  $\rho_{ij}$  is its magnitude.

**Visibility under operator error.** If  $\tilde{A}' = \tilde{A} + \Delta$ , the triangle inequality gives  $|\|\tilde{\mathbf{a}}'_j\|_2 - \|\tilde{\mathbf{a}}_j\|_2| \leq \|\Delta \mathbf{e}_j\|_2$ , so the weak set is unchanged when  $|v_j - \tau_v| > \|\Delta \mathbf{e}_j\|_2$  for every  $j$ .

To express the same scale-invariant geometry as a distance, we also report the ray distance for eligible, generally signed projected fingerprints,  $d_{ij}^{\text{ray}} = \min_{\alpha \in \mathbb{R}} \|\tilde{\mathbf{a}}_i - \alpha \tilde{\mathbf{a}}_j\|_2 / \|\tilde{\mathbf{a}}_i\|_2$ . For nonweak fingerprints,

$$d_{ij}^{\text{ray}} = \sqrt{1 - \rho_{ij}^2}, \quad (9)$$

which we derive in Appendix A; numerical implementations evaluate  $\sqrt{\max(0, 1 - \rho_{ij}^2)}$ . This is an unoriented linear-ray diagnostic, not a claim that projected fingerprints are nonnegative. High coherence and small ray distance encode the same pairwise geometry, so ray distance is a diagnostic geometrically equivalent to coherence under this normalization, not an independent identifiability criterion. Like coherence, it is “N/A” for pairs involving  $\mathcal{W}$  and excluded from distance extrema.

**Background absorption.** Background correction can improve robustness by removing broad temporal variation, but it can also remove source-driven signal. Let  $P_Q = QQ^{\dagger}$ . For coefficient fingerprint  $j = (k, b)$ , define the background absorption fraction  $a_j = \|P_Q \mathbf{a}_j\|_2 / \|\mathbf{a}_j\|_2$ . If  $\|\mathbf{a}_j\|_2$  is zero at numerical tolerance,  $a_j$  is undefined and reported as N/A; the coefficient remains explicitly weak. If  $a_j \approx 1$ , most of the raw source-basis fingerprint lies in the background space, and the projected visibility  $v_j$  will be small even if the raw fingerprint is large. This is why identifiability must be assessed after projection.

## C UNCERTAINTY, ADEQUACY, AND REPORTING DETAILS

This appendix expands the reporting components summarized in Section 5, and gives the IASA thresholds (Table 3).

**Uncertainty reporting.** Let  $\tilde{r} = \tilde{Y} - \tilde{A}\hat{\mathbf{c}}$  be the projected residual and  $\hat{\sigma}^2 = \|\tilde{r}\|_2^2 / \max(N - r_{\text{eff}}(\tau_{\sigma}), 1)$  a

residual variance estimate. For an unconstrained or active-set approximation, let  $\mathcal{A}_c = \{j : \hat{c}_j > \tau_c\}$  and let  $\tilde{A}_{\mathcal{A}_c}$  contain the active columns. The covariance of the active estimates is approximated by

$$\text{Cov}(\hat{\mathbf{c}}_{\mathcal{A}_c}) = \hat{\sigma}^2(\tilde{A}_{\mathcal{A}_c}^\top \tilde{A}_{\mathcal{A}_c})^\dagger. \quad (10)$$

The use of  $N - r_{\text{eff}}(\tau_\sigma)$  is a heuristic degrees-of-freedom count that does not exactly account for the background rank  $r$  or the NNLS active-set constraints. Equation 10 is the standard active-set least-squares covariance approximation (Seber and Lee, 2003): its diagonal entries give per-coefficient uncertainty and its off-diagonal entries expose coupled (jointly uncertain) estimates. For each transport ensemble member from Appendix D.1, IASA rebuilds  $\tilde{A}^{(b)}$ , refits  $\hat{\mathbf{c}}^{(b)}$ , and reports empirical quantiles across members as transport uncertainty. Every ensemble is tagged by provenance: wind and dispersion members have `ensemble_kind=transport` and may form transport-uncertainty intervals, whereas alternative inventory maps, source locations, labels, or scales have `ensemble_kind=inventory` and produce scenario-wise robustness tables. The two kinds are never pooled into one interval, and active-set intervals describe the fitted observation model without converting inventory scenarios into probabilistic uncertainty.

**Residual model-adequacy check.** Let  $Z$  be an orthonormal basis for the background-orthogonal observed-row space, and define the nondegenerate residual coordinates and externally supplied noise covariance

$$\bar{r} = Z^\top (Y - A\hat{\mathbf{c}}), \quad \bar{\Sigma}_e = Z^\top \Sigma_e Z, \quad (11)$$

where  $\Sigma_e$  is an externally declared observation-noise covariance and  $Z$  removes the background directions so that  $\bar{r}$  carries no degrees of freedom absorbed by  $Q$ . For independent homoscedastic noise  $\Sigma_e = \sigma_e^2 I$ , orthonormality of  $Z$  gives  $\bar{\Sigma}_e = \sigma_e^2 I$  and  $T_{\text{res}} = \|\bar{r}\|_2^2 / \sigma_e^2$ . When  $\bar{\Sigma}_e$  is calibrated independently of this fitted residual and is positive definite, the adequacy statistic is

$$T_{\text{res}} = \bar{r}^\top \bar{\Sigma}_e^{-1} \bar{r}. \quad (12)$$

Its null distribution is not a plain  $\chi^2$ , because  $\hat{\mathbf{c}}$  is fitted from the same data; IASA therefore calibrates it by parametric bootstrap. It simulates  $B_{\text{res}} = 1000$  synthetic observation sets from the fitted source/background model plus the declared noise model, refits the coefficients on each draw, and recomputes  $T_{\text{res}}$  to form the null distribution. At  $\alpha = 0.05$  it flags inadequacy when the observed statistic exceeds the empirical  $(1 - \alpha)$ -quantile; the Monte Carlo  $p$ -value uses the standard add-one correction. If no externally calibrated noise model is supplied, IASA reports raw and projected residual norms and sensor-wise, time-wise, and autocorrelation summaries with `calibration_status=uncalibrated` and emits no pass/fail adequacy claim. Rejection establishes a mismatch with the declared model but does not identify a missing source, and non-rejection does not establish inventory completeness: an omitted source whose sensor-time signature lies in  $\text{span}[A, Q]$  can be absorbed by fitted terms and remain residual-invisible.

**Wind-distribution diagnostics.** Diagnostics for one realized wind window answer a retrospective question. Prospective network adequacy is assessed empirically over predeclared historical or simulated wind windows. Across those response matrices, IASA reports the 5th, 50th, and 95th percentiles of  $\sigma_J$ , the probability of full numerical and effective rank, the probability that each coefficient is weak, and the frequency of each grouping  $\mathcal{P}^* \vee \mathcal{D}$ . These are Monte Carlo or empirical distributional summaries under the sampled wind population, not a new identifiability theorem.

**Acceptance criteria for refinement.** If the optional constrained refinement of Appendix D.3 is used, it is accepted only if it improves fit without degrading separability. Let  $\tilde{A}_0$  be the projected response before refinement and  $\tilde{A}_{\text{ref}}$  after refinement. We require  $\sigma_J(\tilde{A}_{\text{ref}}) \geq (1 - \eta_{\text{id}})\sigma_J(\tilde{A}_0)$  and that the grouping  $\mathcal{P}^* \vee \mathcal{D}$  is unchanged. These checks prevent end-to-end tuning from improving sensor fit by making source-basis fingerprints less distinguishable. If either response has fewer than  $J$  numerically nonzero singular values, its  $\sigma_J$  is taken to be zero (Section 5), so refinement cannot be accepted by exploiting a rank-deficient response.

**Reporting protocol.** The atoms are the blocks of  $\mathcal{P}^* \vee \mathcal{D}$ , with  $\mathcal{P}^*$  computed at  $\tau_{\text{num}}$  and  $\mathcal{D}$  the declared source partition. The reported partition is the partition into unions of atoms with every block’s separation at least  $\tau_\theta$  that has the most blocks, with ties broken by the largest smallest separation and remaining ties listed (Section 4). For each block  $G$ , IASA reports the group signal  $\tilde{A}_G \hat{\mathbf{c}}_G$ , its  $L_1$  share, its separation  $s_G$ , the bound  $b_G$  of Theorem 5.9(c) at the declared noise level, the member visibilities and, for a source with several basis columns, the activity trajectory  $\hat{\theta}_{G,1:T}$ . Coefficients are reported only for blocks with one component. Weak coefficients keep their basis names in the report, and coherences involving weak fingerprints are displayed as “N/A”. The

recorded runs computed their groupings with the pairwise coherence rule at 0.99 (merge two declared blocks when a pair of their nonweak columns exceeds it); on every saved operator and observed week this gives the same partition as the separation rule.

## D APPLICATION: CITY-SCALE SOURCE ATTRIBUTION

This appendix gives the construction of the operator, the nuisance basis and the platform for the pollution application summarized in Section 3. In the notation of the general model,  $A(\omega) = H_{\Phi}^{\text{lag}}$ ,  $\hat{A} = \hat{H}_{\Phi}$ , the columns  $\mathbf{h}_{kb}$  are the fingerprints  $\mathbf{a}_j$  with  $j = (k, b)$ , and the background basis  $Q$  is the nuisance basis.

### D.1 Transport Model, Inventories and Lagged Response

**City-scale transport model.** Let  $u(\mathbf{x}, t)$  be the pollutant concentration field on  $\Omega \subset \mathbb{R}^d$ , evolving as  $\partial u / \partial t = \mathcal{F}_{\eta}(u, \mathbf{x}, t; w(t)) + s(\mathbf{x}, t) + b(\mathbf{x}, t)$  under meteorological state  $w(t)$ , local emission  $s$ , background  $b$ , and transport/dispersion parameters  $\eta$ ; the dynamics may be realized by a plume model, an advection–diffusion solver, or a transport-constrained surrogate.

**Sparse observations.** Discretize the concentration field on  $n$  grid cells, so that  $\mathbf{u}_t \in \mathbb{R}^n$ . Let  $X_{\text{sens}} = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$  be the fixed sensor locations with  $m \ll n$ . The observation operator  $O : \mathbb{R}^n \rightarrow \mathbb{R}^m$  samples or interpolates the latent field at these locations,  $\mathbf{y}_t = O\mathbf{u}_t + \epsilon_t$ ; stacking over  $t = 1, \dots, T$  gives  $Y = [\mathbf{y}_1^{\top}, \dots, \mathbf{y}_T^{\top}]^{\top} \in \mathbb{R}^{mT}$ . Because  $m$  is small relative to the field dimension, many latent fields and source configurations fit the same sensor trajectory.

**Inventory-based source parameterization.** We assume a set of  $K$  known or proxy source maps,  $S = [\mathbf{s}_1 \dots \mathbf{s}_K] \in \mathbb{R}^{q \times K}$ , where  $\mathbf{s}_k \in \mathbb{R}^q$  is the spatial pattern of source group  $k$ . Time-varying source activity uses a fixed, predeclared nonnegative temporal dictionary  $\Phi = [\phi_1 \dots \phi_B] \in \mathbb{R}_+^{T \times B}$  and coefficients  $C = [c_{kb}] \in \mathbb{R}_+^{K \times B}$ , so the activity of source  $k$  is  $\theta_k(t) = \sum_{b=1}^B c_{kb} \phi_b(t) \geq 0$ . The basis is a shared dictionary of admissible temporal patterns, not a claim that all sources follow one trajectory: each source has its own nonnegative weights, and source-specific knowledge is imposed by masking inadmissible coefficients. We do not learn  $\Phi$  from the same sparse observations, since jointly estimating  $\Phi$  and  $C$  would add bilinear non-identifiability beyond the source distinguishability analyzed here. We vectorize  $C$  in source-major, basis-minor order as  $\mathbf{c} = \text{vec}_{\text{src}}(C) \in \mathbb{R}_+^J$ ,  $J = KB$ ; the constant-activity model is the special case  $B = 1$ ,  $\phi_1(t) = 1$ .

**Wind-conditioned lagged response.** Government wind direction records where wind comes from, whereas transport needs the direction pollution moves; we convert direction  $WD$  and speed  $WS$  to transport vectors

$$(U_x, V_y) = -WS (\sin \alpha, \cos \alpha), \quad \alpha = WD \pi / 180, \quad (13)$$

and complete them on the response grid with a normalized Gaussian-kernel coordinate-query imputer to obtain a gridded transport field  $\hat{\mathbf{w}}_t(\mathbf{x}_g)$ ; and the same interface supplies synthetic controlled winds and transport ensembles kept separate from inventory scenarios. Emissions released at  $t - \ell$  affect sensors at  $t$  only after transport by the intervening wind. Let  $G_{t, t-\ell}^{\partial}(w_{t-\ell:t}) \in \mathbb{R}^{n \times q}$  map emissions released on the source grid at  $t - \ell$  to the concentration grid at  $t$ ; the superscript  $\partial$  denotes an open-boundary operator (mass leaving  $\Omega$  is removed, not reflected or wrapped). It is realized by a Gaussian puff approximation evaluated at sensor coordinates that preserves nonnegativity and is mass non-increasing inside the domain, with dispersion, exit bookkeeping, and the pre-fit initial-condition baseline given in Appendix D.3. For a maximum lag  $L$ ,  $\mathcal{L}_t = \{0, \dots, \min(L, t - 1)\}$ , the unit-coefficient sensor fingerprint of source  $k$  and basis component  $b$  at time  $t$  is

$$\mathbf{h}_{t, kb}^{\text{lag}} = \sum_{\ell \in \mathcal{L}_t} \phi_b(t - \ell) O G_{t, t-\ell}^{\partial}(w_{t-\ell:t}) \mathbf{s}_k \in \mathbb{R}^m. \quad (14)$$

Stacking over time in source-major, basis-minor order gives  $H_{\Phi}^{\text{lag}} \in \mathbb{R}^{mT \times J}$ ,  $J = KB$ , each column the finite-horizon sensor-time pattern produced by one unit of a coefficient. The lag window adds no sensor–time rows and is declared before fitting ( $L = 12$  hours in every reported run except the lag sweep of Experiment 7). A convergence diagnostic compares adjacent candidates  $L$ ,  $L + \Delta$  with the same rows and columns,

$$\eta_L = \frac{\|H_{\Phi}^{\text{lag}}(L + \Delta) - H_{\Phi}^{\text{lag}}(L)\|_F}{\max\{\|H_{\Phi}^{\text{lag}}(L + \Delta)\|_F, \epsilon\}}, \quad (15)$$

and would select the smallest candidate with  $\eta_L \leq \tau_L$  (default  $10^{-3}$ ); on the grid of Experiment 7 no candidate meets it. The fitted  $\mathbf{c}$  never selects  $L$ .

**Background correction and projection.** Observed concentrations include components not explained by the local inventory (regional background, smooth trends, sensor offsets), giving  $Y = H_\Phi^{\text{lag}} \mathbf{c} + Q\gamma + E$  with a low-dimensional basis  $Q \in \mathbb{R}^{mT \times r}$  specified before fitting from source-independent metadata only (timestamps, day labels, sensor identity, and coordinates; effective rank capped at eight; rank four in every reported run except the nuisance-basis variations of Experiments 3 and 11), so apportionment is not confounded by background correction; columns built from inventories, fingerprints, or fitted signals are excluded and permitted only in labeled stress tests (Appendix D.3). Real records need not be complete: let  $\mathcal{O}$  be the ordered observed sensor–time  $\text{PM}_{2.5}$  rows and  $M_{\mathcal{O}} \in \{0, 1\}^{N \times mT}$  select them, applied identically to every row-aligned object,

$$Y_{\mathcal{O}} = M_{\mathcal{O}}Y, \quad H_{\Phi, \mathcal{O}}^{\text{lag}} = M_{\mathcal{O}}H_\Phi^{\text{lag}}, \quad Q_{\mathcal{O}} = M_{\mathcal{O}}Q, \quad (16)$$

with an alignment mismatch treated as an error; wind may be imputed,  $\text{PM}_{2.5}$  is not, and  $M_{\mathcal{O}} = I$  for complete controlled data. Dropping the  $\mathcal{O}$  subscript, let  $P_Q^\perp = I - QQ^\dagger$  project onto the orthogonal complement of the background space (applied implicitly through a thin SVD of  $Q$ ); the corrected inverse problem is

$$\tilde{Y} = \tilde{H}_\Phi \mathbf{c} + \tilde{E}, \quad \tilde{Y} = P_Q^\perp Y, \quad \tilde{H}_\Phi = P_Q^\perp H_\Phi^{\text{lag}}, \quad (17)$$

so each projected fingerprint  $\tilde{\mathbf{h}}_{kb} = P_Q^\perp \mathbf{h}_{kb}^{\text{lag}}$  encodes the observable effect of one source–basis coefficient after transport, lag, sparse sensing, and background correction. All identifiability diagnostics in this paper are computed from  $\tilde{H}_\Phi$ , not the raw inventory or the unprojected response.

## D.2 Temporal-Basis Notation and Diagnostic Conventions

Table 4: Indices and objects in the temporal-basis apportionment model.

| Symbol                    | Shape or range | Meaning                                     |
|---------------------------|----------------|---------------------------------------------|
| $k$                       | $1, \dots, K$  | source-group index                          |
| $b$                       | $1, \dots, B$  | temporal-basis index                        |
| $j = (k, b)$              | $1, \dots, J$  | source-major, basis-minor coefficient index |
| $\Phi$                    | $T \times B$   | nonnegative source-activity basis           |
| $C$                       | $K \times B$   | nonnegative source–basis coefficients       |
| $\mathbf{c}$              | $J = KB$       | source-major vectorization of $C$           |
| $H_\Phi^{\text{lag}}$     | $N \times J$   | observed-row coefficient fingerprints       |
| $\tilde{H}_\Phi$          | $N \times J$   | background-projected fingerprints           |
| $\tilde{\mathbf{h}}_{kb}$ | $N$            | fingerprint for coefficient $c_{kb}$        |

The constant-activity model uses  $B = 1$ ,  $\phi_1(t) = 1$ , and  $c_{k1} = \theta_k$ . In the general model, diagnostics are computed for the  $J$  coefficient fingerprints. Source-level activity trajectories are reconstructed as

$$\hat{\theta}_k(t) = \sum_{b=1}^B \hat{c}_{kb} \phi_b(t);$$

fingerprint columns are not summed to create an artificial source-level diagnostic vector.

For a source-level ambiguity summary, IASA reports the largest eligible coherence between coefficients belonging to different sources and retains the source–basis pair that attains it. Coherence and ray distance are undefined for pairs involving a coefficient with  $v_j \leq \tau_v$ ; tables display these entries as N/A, exclude them from pairwise extrema, and report the weak coefficient separately.

## D.3 Transport Response Construction and Fit Details

This appendix gives the construction, background, and estimation details deferred from Section 3.

**Wind imputer query and unit conversion.** The observed station vectors and masks are supplied to a coordinate-query wind imputer: given sparse station observations  $\{(\mathbf{z}_i, t, \mathbf{u}_{i,t}, m_{i,t})\}$ , it predicts transport vectors

at arbitrary spatial query locations and times. For the IASA response operator, we query the imputer on every response-grid cell  $\mathbf{x}_g$  and hour  $t$ , producing

$$\widehat{\mathbf{w}}_t(\mathbf{x}_g) = \widehat{f}_\omega(\mathbf{x}_g, t; \{(\mathbf{z}_i, t', \mathbf{u}_{i,t'}, m_{i,t'})\}), \quad g = 1, \dots, n. \quad (18)$$

This yields a gridded wind field  $\widehat{W} \in \mathbb{R}^{T \times n \times 2}$ , rather than a station-only or city-averaged sequence. Real-data validation is performed by masking observed station vectors and measuring held-out station-time error; dense city-wide wind truth is not assumed. Meteorological speed must be converted to grid displacement before puff advection. For response timestep  $\Delta t_s$  seconds and cell dimensions  $\Delta x_m, \Delta y_m$ , we use

$$v_x^{\text{grid}}(\mathbf{x}_g, t) = \frac{\widehat{U}_x(\mathbf{x}_g, t) \Delta t_s}{\Delta x_m}, \quad v_y^{\text{grid}}(\mathbf{x}_g, t) = \frac{\widehat{V}_y(\mathbf{x}_g, t) \Delta t_s}{\Delta y_m}. \quad (19)$$

Every response artifact records these scales, the coordinate convention, the wind units, and the exact converted sequence.

**Wind-field ensembles.** The imputed gridded wind field is an estimated transport input, not ground truth. To propagate wind-imputation uncertainty, we construct wind-field ensembles  $\{\widehat{\mathbf{w}}_t^{(r)}(\mathbf{x}_g)\}_{r=1}^R$  using held-out-calibrated prediction error, station bootstrap resampling, checkpoint ensembles, or perturbations matched to validation residuals. Each ensemble member is queried on the same response grid and converted to grid displacement before response construction. The resulting response matrices are tagged as transport ensembles and are kept separate from inventory-robustness scenarios.

**Puff dynamics and dispersion.** We instantiate the response with a Gaussian puff approximation. For a unit release from source cell  $i$  at location  $\mathbf{r}_i$  and release time  $\tau$ , the puff center is initialized as  $\mathbf{z}_i(\tau) = \mathbf{r}_i$  and advected by the interpolated wind field,  $d\mathbf{z}_i(a)/da = \widehat{\mathbf{w}}_a(\mathbf{z}_i(a))$ ,  $a \in [\tau, t]$ . Source cells have integer centers and domain extents  $[-0.5, N_x - 0.5] \times [-0.5, N_y - 0.5]$ . With substep size  $\delta a$ , the trajectory is discretized as  $\mathbf{z}_i^{r+1} = \mathbf{z}_i^r + \delta a \widehat{\mathbf{w}}_{a_r}(\mathbf{z}_i^r)$ . The final substep is shortened so that the trajectory lands exactly at the requested observation time. If the center exits  $\Omega$ , the whole remaining puff is removed and never reflected, wrapped, clamped, or reinserted. Otherwise its contribution is spread with an anisotropic Gaussian kernel

$$K_\psi(\mathbf{x}; \mathbf{z}_i, \Sigma_i) = \frac{\exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{z}_i\|_{\Sigma_i^{-1}}^2\right)}{2\pi|\Sigma_i|^{1/2}}, \quad (20)$$

where  $\psi$  denotes dispersion parameters. A simple covariance model is

$$\Sigma_i(t, \tau) = R_i(t, \tau) \begin{bmatrix} \sigma_{\parallel}^2(t - \tau) & 0 \\ 0 & \sigma_{\perp}^2(t - \tau) \end{bmatrix} R_i(t, \tau)^\top, \quad (21)$$

with  $R_i$  aligned to the mean wind direction along the trajectory. When the mean wind norm is below the numerical threshold, the downwind direction is undefined; the implementation uses the positive- $x$  axis as a deterministic tie-breaker for the anisotropic covariance orientation, recorded in the response metadata and affecting only near-calm releases. The age in this covariance is lower-bounded by a positive minimum dispersion time, so same-time releases remain finite: effective age =  $\max(t - \tau, t_{\min})$ .

**Sensor evaluation and loss bookkeeping.** Response-matrix entries are computed by evaluating the kernel directly at sensor coordinates:

$$[OG_{t,\tau}^\partial(\widehat{\mathbf{w}}_{\tau:t}; \psi) \mathbf{e}_i]_s = \chi_i(t, \tau) K_\psi(\mathbf{x}_s; \mathbf{z}_i(t), \Sigma_i(t, \tau)). \quad (22)$$

Here  $\chi_i(t, \tau)$  is an open-boundary survival indicator for a release from source cell  $i$  at release time  $\tau$ : it is one while the advected kernel center remains inside the modeled domain and zero after the release has exited. Boundary and truncation losses are recorded only as diagnostic quantities and are never used to renormalize the sensor response. Boundary loss denotes the portion of the Gaussian kernel lying outside the modeled spatial domain, while truncation loss denotes the portion omitted by evaluating retained mass only over a finite kernel support; both are estimated separately by grid quadrature over the response grid. These kernel losses are distinct from whole-release exit loss, where the advected release center leaves the open domain and no later sensor contribution is assigned to that release. Because the response matrix evaluates concentration at sensor points rather than

integrating over area-weighted grid cells, summing entries across sensors or repeated sensor-time observations does not estimate retained pollutant mass.

**Baseline policy and differentiability.** Response matrices are constructed for source-induced increments relative to a declared initial-condition baseline. The baseline policy is fixed before source fitting, recorded with the response artifact, and applied consistently to the observations and response rows. Application-specific choices, such as whether to subtract a zero-source transported initial field or to use a zero-initial-state response, are part of the experimental instantiation rather than the generic response definition. The open-boundary lagged plume response is implemented in PyTorch; tensor operations retained in the graph can be differentiated, but discrete exit events and reporting logic are not assigned gradients.

**Background basis and implicit projection.** The normal background basis may contain a global constant, centered and scaled elapsed-time polynomials, local-clock daily sine/cosine harmonics, reference-coded day effects, standardized regional sensor-coordinate trends, reference-coded sensor offsets, and a row-aligned user basis. It depends only on timestamps, day labels, sensor identity, and sensor coordinates, and its effective rank is capped at eight. Once the primary  $Q$  is fixed, its coefficients are estimated implicitly by projection, or explicitly in the joint fit; each alternative source-independent basis reports the resulting visibility, absorption, rank, conditioning, and ambiguity components. Letting  $Q = U\Sigma V^\top$  be a thin singular value decomposition and retaining columns  $U_r$  above the recorded rank tolerance, projection is applied implicitly,

$$P_Q^\perp X = X - U_r(U_r^\top X), \quad (23)$$

without constructing an  $N \times N$  matrix. The row metadata for  $Y$ ,  $H_\Phi^{\text{lag}}$ , and  $Q$  must be identical: each row must refer to the same timestamp, sensor identity, and sensor index in the same order before projection is applied.

**Regularizers and joint fit.** The implementation also accepts a penalty  $\lambda R(\mathbf{c})$  added to the objective of Equation 4, with ridge stabilization  $R(\mathbf{c}) = \|\mathbf{c}\|_2^2$  or a weak prior around inventory-derived activity levels  $R(\mathbf{c}) = \|\mathbf{c} - \mathbf{c}_0\|_2^2$ ; every reported run uses  $\lambda = 0$ . Equivalently, one may fit background and source coefficients jointly,

$$(\hat{\mathbf{c}}, \hat{\gamma}) = \arg \min_{\mathbf{c} \geq 0, \gamma} \|Y - H_\Phi^{\text{lag}} \mathbf{c} - Q\gamma\|_2^2 + \lambda R(\mathbf{c}). \quad (24)$$

The projected formulation is used for identifiability because it exposes the part of each source-basis fingerprint that remains after background correction.

**Constrained end-to-end refinement.** No reported run uses the refinement described here. IASA first constructs a fixed response matrix from the wind field and default dispersion parameters, fits source activities, and computes identifiability diagnostics for that declared response. It then optionally performs a constrained local refinement of the wind field, dispersion parameters, source coefficients, and background coefficients, initialized at the fixed-response solution and allowed only to make small physically constrained corrections, so that improved sensor fit does not come from arbitrary changes to transport. Let  $\phi$  denote the wind-imputer parameters and  $\psi$  the dispersion parameters of the puff response (such as  $\sigma_\parallel$ ,  $\sigma_\perp$ , and the minimum dispersion age), with  $\phi_0$  and  $\psi_0$  their pretrained or default values. We use  $R_{\text{sm}}$  for a predeclared wind-field smoothness penalty (temporal or spatial squared differences) and  $\Psi_{\text{phys}}$  for the physically admissible dispersion-parameter set. The refinement objective is

$$\begin{aligned} \mathcal{L}_{\text{refine}} = & \|Y - H_\Phi^{\text{lag}}(\phi, \psi) \mathbf{c} - Q\gamma\|_2^2 + \lambda_\theta R(\mathbf{c}) \\ & + \lambda_w \|\hat{\mathbf{w}}_{1:T}^\phi - \hat{\mathbf{w}}_{1:T}^{\phi_0}\|_2^2 + \lambda_\psi \|\psi - \psi_0\|_2^2 \\ & + \lambda_{\text{sm}} R_{\text{sm}}(\hat{\mathbf{w}}^\phi), \end{aligned} \quad (25)$$

subject to  $\mathbf{c} \geq 0$ ,  $\psi \in \Psi_{\text{phys}}$ , and  $\|\hat{\mathbf{w}}_{1:T}^\phi - \hat{\mathbf{w}}_{1:T}^{\phi_0}\|_\infty \leq \epsilon_w$ . The refined response is accepted only as a constrained local correction (acceptance criteria in Appendix C) and is reported with its own response diagnostics. Without these restrictions, wind and plume parameters could fit sparse pollution sensors while weakening the physical and source-level interpretation of  $\hat{\mathbf{c}}$  and the reconstructed activities.

#### D.4 Response-Matrix Perturbations

This appendix expands the response-error discussion of Section 5. Write the projected response discrepancy as  $\Delta_H = \Delta_{\text{tr}} + \Delta_{\text{inv}} + \Delta_{\text{int}}$ , where  $\Delta_{\text{tr}}$  is the transport term,  $\Delta_{\text{inv}}$  the inventory term, and  $\Delta_{\text{int}}$  their interaction.

Table 5: New Delhi source groups, spatial proxies, and temporal-basis components.

| Source group       | Spatial proxy         | Temporal-basis components    |
|--------------------|-----------------------|------------------------------|
| Brick kilns        | regional kiln map     | alternating 12-hour blocks   |
| Industries         | regional industry map | continuous + daytime (07–19) |
| Population density | gridded population    | peaks 07:00, 13:00, 19:00    |
| Traffic            | road-network map      | diurnal slots 00/06/12/18    |

The decomposition keeps the two error sources apart:  $\Delta_{\text{tr}}$  is propagated through transport ensembles and may become an uncertainty interval, whereas  $\Delta_{\text{inv}}$  is reported only as named robustness scenarios and is never pooled into an interval, and  $\Delta_{\text{int}}$  is recorded rather than pooled separately. Let  $\tilde{H}_{\Phi}^*$  be the true projected response and  $\hat{H}_{\Phi}$  the constructed one, with  $\|\hat{H}_{\Phi} - \tilde{H}_{\Phi}^*\|_2 \leq \delta_H$ . Then  $\tilde{Y} = \hat{H}_{\Phi}\mathbf{c} + [\tilde{E} + (\tilde{H}_{\Phi}^* - \hat{H}_{\Phi})\mathbf{c}]$ , so response error acts like additional observation error.

**Proposition D.1** (Perturbation bound). *Assume  $\hat{H}_{\Phi}$  has full column rank and  $\|\hat{H}_{\Phi} - \tilde{H}_{\Phi}^*\|_2 \leq \delta_H$ . The unconstrained least-squares estimate using  $\hat{H}_{\Phi}$  satisfies*

$$\|\hat{\mathbf{c}} - \mathbf{c}\|_2 \leq \frac{\|\tilde{E}\|_2 + \delta_H \|\mathbf{c}\|_2}{\sigma_J(\hat{H}_{\Phi})}. \quad (26)$$

A short proof, reusing the pseudoinverse bound of Proposition 5.4, appears in Appendix A. The same norm bound applies to all three discrepancy terms, but their interpretations are not interchangeable. Wind and dispersion ensembles quantify transport uncertainty and may support uncertainty intervals under the ensemble model. Alternative inventory maps, locations, labels, or scales are inventory robustness scenarios, not confidence intervals. They are reported separately, and source-specific attribution remains conditional on the supplied inventory. By Proposition D.1, either error type increases the coefficient error more when  $\sigma_J$  is small.

## D.5 Per-Sensor Source Footprints

Policy questions are often also spatial, and because the response is linear in  $\mathbf{c}$ , the fitted solution already localizes each monitor’s signal. For an observed row  $(s, t)$ , the background-corrected fit decomposes exactly as  $\tilde{y}_{s,t} = \sum_{k,b} \tilde{H}_{\Phi,(s,t),(k,b)} \hat{c}_{kb}$ , so the contribution of source  $k$  to sensor  $s$  over the record is  $\hat{Y}_k^{(s)} = \sum_t \sum_b H_{\Phi,(s,t),(k,b)}^{\text{lag}} \hat{c}_{kb}$ . The projected form isolates the identifiable part; the unprojected form gives the raw fitted sensor contribution before background removal. The spatial origin of a sensor’s signal is the pullback of its response row onto the source grid: for sensor  $s$  at time  $t$ ,  $F_{s,t}(i) = \sum_{\ell \in \mathcal{L}_t} [OG_{t,t-\ell}^{\partial}(\hat{\mathbf{w}}_{t-\ell:t})]_{s,i} \geq 0$  and  $F_s(i) = \sum_t F_{s,t}(i)$ , a nonnegative field over source cells  $i$  recording which upwind cells influence sensor  $s$ . Weighting by  $\phi_b(t - \ell)$ , the inventory map  $\mathbf{s}_k$ , and  $\hat{c}_{kb}$  resolves the fitted contribution of each source group to each sensor by cell of origin. This is the puff response of Appendix D.1 read backward from the sensor and requires no new transport operator. Per-sensor attribution uses only the rows of  $\tilde{H}_{\Phi}$  for that sensor: for the row submatrix  $\tilde{H}_{\Phi}^{(s)}$ ,  $\tilde{H}_{\Phi}^{(s)\top} \tilde{H}_{\Phi}^{(s)} \preceq \tilde{H}_{\Phi}^{\top} \tilde{H}_{\Phi}$ , so deleting rows cannot increase the smallest singular value or the rank and a single sensor is never more identifiable than the pooled network. Footprints and per-sensor contributions are therefore reported as spatial overlays of the global fit; per-sensor source *shares* are asserted only at the globally identifiable resolution, aggregating to a report group wherever the pooled diagnostics require one, which keeps the spatial view from implying separations the sensing system does not support.

## D.6 New Delhi Platform Details

This appendix expands the platform construction of Appendix E: the source inventories (Table 5), the kriged initial-condition baseline, and the auxiliary simulator.

**Observations, sensors, and inventory preparation.** The hourly government records (CPCB, 2023, 2025) (columns `monitor_id`, `timestamp_round`, `AT`, `RH`, `WD`, `WS`, `pm10`, `pm25`) span 21,960 hourly timestamps from 1 May 2018 through 31 October 2020 in Indian Standard Time; over all 32 locations, including East Arjun Nagar, which has no readings, approximately 74.9% of wind-direction, 72.8% of wind-speed, and 90.5% of  $\text{PM}_{2.5}$  entries are observed, with a valid-value mask retained alongside station coordinates and timestamps. The two Pusa monitors

(Pusa\_IMD and Pusa\_DPCC) are averaged by timestamp into `Pusa_averaged` (mean coordinates, per-variable hourly averaging), giving the final 32-sensor layout;  $\text{PM}_{2.5}$  is never imputed and its flattened valid-value mask defines  $M_{\mathcal{O}}$ , with the same ordered sensor–time rows retained in  $Y$ ,  $H_{\Phi}^{\text{lag}}$ ,  $Q$ , and metadata. Brick-kiln and industry proxies follow the regional emissions inventory (Guttikunda and Calori, 2013), population density the gridded population product (Center for International Earth Science Information Network (CIESIN), Columbia University, 2018), and the traffic map is derived from road map data (Google, 2024). Every  $80 \times 80$  map is cropped to the New Delhi study window (rows 21:61, columns 16:56) covering the regulatory network, giving a  $40 \times 40$  map, then divided by its own cropped 99th percentile. Traffic is modeled as a single road-network source whose nearest-slot maps at 00:00, 06:00, 12:00, and 18:00 define traffic-specific temporal-basis components under the shared dictionary  $\Phi$  and the fixed-zero mask  $\mathcal{F}_0$  (assuming a time-invariant congestion spatial pattern whose amplitude varies); brick-kiln activity uses deterministic alternating 12-hour blocks, industry a continuous operating fraction with higher daytime (07:00–19:00) activity, and population a baseline with smooth cooking peaks. These are deterministic study assumptions, not observed emissions truth, and each map’s independent proxy normalization is why fitted coefficients are in normalized proxy units; the group signals are in  $\mu\text{g}/\text{m}^3$ , and the shares are not emission shares. The  $40 \times 40$  crop is a fixed index window of the inventory grid (rows 21–60, columns 16–55); the code does not derive it from the sensor coordinates, and its alignment with the bounding box of the sensors, which defines the sensor and wind grid, has not been verified.

**Forward-model baselines and study modes.** Both forward paths use the New Delhi-derived  $40 \times 40$  domain and regulatory sensor coordinates, with the open-boundary Gaussian puff operator evaluated directly at sensors. The same operator is the matched generator for controlled recovery experiments and the inference operator for observed  $\text{PM}_{2.5}$ ; only the initial-condition baseline differs by mode: a zero-source, zero-initial-state baseline for controlled operator-matched runs, and a subtracted zero-source transported kriged-initial-condition baseline for observed runs. Normal backgrounds follow Appendix D.1, with the base controlled background the rank-four member. Where a study is explicitly about inventory geometry (the coherence, inventory-robustness and missing-source experiments) the real New Delhi inventory maps are used; the remaining controlled experiments use compact synthetic source cells for tractability, since their conclusions concern conditioning, coherence, and recovery rather than specific inventory footprints.

**Kriged initial-condition baseline.** For observed New Delhi runs, the initial pollution field is estimated from the first hour of the window in which at least three stations report, by a normalized Gaussian-kernel spatial interpolation on the response grid (a kriging surrogate). Its zero-source transported trajectory, obtained by propagating this initial field as a single release through the same open-boundary puff operator (its Gaussian kernel is the advection–diffusion Green’s function, so units are preserved), is subtracted from the sensor observations before fitting source activities. Source coefficients are therefore fitted to the residual sensor-time signal after removing the contribution of the initialized background field and the low-rank temporal background basis. Kernel interpolation from sparse first-hour observations is itself an ill-posed spatial estimate, so this baseline injects some uncertainty; the effect is limited by declaring and subtracting the baseline before fitting and by the low-rank background basis  $Q$ , which absorbs residual broad structure. Controlled operator-matched experiments may instead use a zero initial field when the synthetic data are generated from the same zero-initial-state response.

**Auxiliary advection–diffusion simulator.** The auxiliary advection–diffusion simulator uses first-order upwind advection, a five-point Laplacian, a two-stage Heun update, diffusivity  $3 \times 10^{-4}$ , and edge-hold boundaries. It provides a structural forward-model mismatch: data generated by this distinct transport operator are fit with the open-boundary puff response, testing the puff approximation against a different operator family. This mismatch also drives the residual-adequacy test of Section 5, letting us check its power against structural transport misspecification and not only against missing sources.

## D.7 Wind-Imputation Validation

The learned wind model did not beat a non-spatial city-mean baseline on held-out station–time vectors, so the spatially resolved kernel imputer is used throughout (the city mean lacks the spatial resolution the response operator requires). Dense wind truth is unavailable, so gridded-field accuracy is assessed only through the controlled synthetic-wind experiments, where the true wind is known.



## E CONTROLLED AND OBSERVED STUDIES: FULL DESCRIPTION

We evaluate on a single New Delhi platform so the controlled and observed studies share one regulatory sensor geometry, one  $\text{PM}_{2.5}$  and wind record, and one set of proxy inventories. We ask whether the geometry of  $\tilde{H}_\Phi$  predicts which coefficient, activity, and grouped-source claims hold. A shared base configuration is varied along one controlled axis at a time, avoiding conflation of wind, source geometry, background flexibility, noise, and forward-model error. All runs use the  $40 \times 40$  resolution, with a 72-hour transport window for the controlled experiments and four consecutive one-week windows for the observed study. Real imputed wind is a controlled transport input when coefficients are synthetically known; the observed- $\text{PM}_{2.5}$  mode is a separate evaluation and is never assigned synthetic source-recovery metrics.

**New Delhi Experimental Platform.** The platform is built from hourly government  $\text{PM}_{2.5}$  and wind records (CPCB, 2023, 2025) spanning 1 May 2018–31 October 2020 over a 32-sensor regulatory network.  $\text{PM}_{2.5}$  is never imputed, and its valid-value mask defines  $M_O$  in Equation 16, with the same ordered sensor–time rows retained in  $Y$ ,  $H_\Phi^{\text{lag}}$ ,  $Q$ , and metadata. Observed wind is converted (Equation 13) and completed on the response grid by the kernel coordinate-query imputer of Appendix D.1. A learned imputer was trained but not adopted (Appendix D.7). Four source groups (brick kilns, industries, population density, traffic) each carry one normalized spatial proxy map and its own admissible temporal-basis components (Table 5, Appendix D.6), so fitted coefficients are in normalized proxy units, not physical emission shares. Both forward paths use the open-boundary Gaussian puff operator of Appendix D.1 evaluated at sensors, the matched generator for controlled recovery and the inference operator for observed  $\text{PM}_{2.5}$ . An auxiliary advection–diffusion simulator supplies a structural forward-model mismatch for the adequacy study (Section 5), and normal backgrounds follow Appendix D.1.

The *controlled* mode preserves this sensor geometry and transport while assigning known nonnegative source–basis coefficients (synthetic or real gridded wind, controlled observation noise, source splits, sensor-layout variants, and optional operator mismatch), so it supports recovery metrics. The *observed* mode combines real  $\text{PM}_{2.5}$ , its observation mask, the imputed wind field, and the named normalized inventories; it has no ground-truth source activities and is run over four consecutive one-week windows, each fitted and reported independently. Column-level records, Pusa-monitor averaging and coverage, proxy normalization and crop, temporal-activity assumptions, forward-model baselines, and simulator settings are given in Appendix D.6.

**Metrics and Reporting.** Controlled trials, having known coefficients, report coefficient and reconstructed-activity error, grouped activity/contribution error and projected residual; interval coverage is not reported. All trials also report the singular spectrum and its diagnostics ( $\sigma_J$ , numerical and effective rank, condition number, visibility, background absorption, eligible coherence and ray distance), the weak set, triggering pairs, and deterministic report groups (Section 5). They also record lag convergence, the fixed-zero mask, background specification, and ensemble provenance (transport and inventory kept separate). Observed New Delhi runs instead report raw and projected residuals, fitted trajectories, normalized proxy coefficients, group signals in  $\mu\text{g}/\text{m}^3$ , uncertainty intervals, inventory-scenario robustness, weak flags, separations, group signals and shares, and residual calibration status. A reported share is an  $L_1$  fraction of the fitted projected group signals, not a physical-emission share.

### E.1 Controlled Results

**Recorded groupings.** The controlled runs recorded groupings from the pairwise rule (merge two declared blocks with a cross-source coherence above 0.99). On every saved operator, and in every observed week, this equals the reportable partition at  $\tau_\theta = 0.141$  with the most blocks (Section 4); for Experiments 3 and 9 it also equals  $\mathcal{P}^* \vee \mathcal{D}$ .

The controlled trials have known source–basis coefficients and therefore recovery metrics. The controlled geometries are small (mostly  $K = 2$  source groups) so that conditioning, coherence, and background absorption can each be varied in isolation and read directly off  $\tilde{H}_\Phi$ . The observed study then exercises the full four-group, seven-coefficient pipeline on real data. Figure 2 collects the baseline comparison and the seven controlled diagnostics. Full numeric tables are in Appendix F.

**Baselines.** We compare IASA with plain NNLS on the projected system (the same estimator without grouping or diagnostics), chemical mass balance on the unprojected response, and a PMF/NMF receptor factorization,

over three seeds (Figure 2(a); Table 1). IASA raises an identifiability flag in every scenario and seed and none of the baselines does. Under background stress the projected source signals stay separated; IASA’s largest group-signal error is 0.028–0.060 against 2.10–4.63 for chemical mass balance, and plain NNLS matches IASA. In the steady-wind case, two seeds have no signal at the sensors; in the third IASA merges the two sources and the merged group’s signal error is 0.004, against 7.84 for chemical mass balance, while the baselines report the individual split without a flag.

**Conditioning and Recovery.** We vary source geometry and Gaussian observation noise (0–20% of the maximum clean sensor signal). Both geometries are full rank (numerical rank = effective rank = 2), so the fixed geometry fixes  $\sigma_J$  and  $\kappa$  and coefficient error grows linearly with noise. The better-conditioned *close* geometry recovers proportionally more accurately than the *separated* one (Figure 2(b); Table 6): at equal noise the coefficient error is set by the response-matrix geometry.

**Wind Diversity and Sensor Geometry.** Matched source, basis, and sensor sets are evaluated across six wind providers (constant, single-direction, diurnal, AR(1), multi-directional, and real gridded New Delhi wind) and three sensor layouts (regulatory, seeded random, downwind-focused), with identical source–basis columns on both sides of every comparison (Figure 2(c); Table 7). Two effects are seen: wind diversity improves conditioning (multi-directional and real wind give the lowest maximum eligible coherence, while a single steady direction pushes it toward one), and the sensor layout changes  $\sigma_J$  as much as the wind does: with the seeded random layout one source is nearly invisible,  $\sigma_J$  is near zero and the only nonzero coefficient errors in the sweep occur, while the regulatory layout keeps  $\sigma_J \geq 1.53$ . Separate historical and AR(1) window ensembles both attain full rank with probability 1.0 and no pairwise ambiguity (Table 8).

**Background Stress.** We compare no background, the normal rank-four basis, a redundant-column basis with the same span, and a labeled source-like stress basis, all declared before fitting. The redundant basis is byte-identical to the primary in every reported quantity, while the source-like *stress* basis drives  $\sigma_J$  and minimum visibility to 0 and absorption to 1 (Figure 2(d); Table 9). A background basis that can reproduce a source’s signal absorbs it and removes its identifiability while the fitted residual is unchanged. The coefficient error *falls* in the stress case because the signal has been absorbed.

**Transport Error.** Parametric perturbations of wind speed, direction, and dispersion within the puff family propagate into attribution. Coefficient error grows with the operator-error norm along the direction (to 0.86 at 20°) and dispersion (to 0.89 at 2×) axes, while the speed axis is non-monotonic, so the operator-error norm does not order the attribution error (Figure 2(e); Table 10). A structural mismatch (open-boundary puff fitted to edge-hold advection–diffusion observations, mismatch norm 0.44) drives mean coefficient error to 0.86 and the refitted-bootstrap adequacy test rejects in 100% of trials, so the test also rejects forward-model misspecification, not only missing sources.

**Lag-Window Selection.** Physical travel and residence times define an increasing candidate lag grid. As the window grows, conditioning improves monotonically ( $\kappa$  falls from 569 at  $L=4$  to 5.1 at  $L=16$ ;  $\sigma_J$  rises from 0.08 to 9.07) while numerical rank and report-component count stay fixed and recovery is exact (Figure 2(f); Table 12). The relative Frobenius changes of Eq. 15 between adjacent candidates are 0.063, 0.096, 0.082, 0.057 and 0.139, so no candidate meets  $\tau_L = 10^{-3}$  and the convergence rule does not select a lag on this grid; the other reported runs use the declared  $L = 12$ .

**Temporal-Basis Recovery.** Reconstructing known diurnal, intermittent, and day/night source–basis coefficients at increasing noise separates coefficient error from reconstructed-activity error. The activity-trajectory error stays consistently below the coefficient error (0.58 vs. 0.90 at 20% noise), at a ratio of 0.57 up to 10% noise and 0.65 at 20% (Figure 2(g); Table 13), because projecting noisy coefficients through the temporal bases damps isotropic estimation noise more than the coherent signal.

**Inventory Robustness.** Perturbing inventory locations, spatial scales, category assignments, and map versions (each a separate scenario, never pooled with transport error) recovers the known coefficients exactly, but the identifiability geometry moves with the inventory version. Rescaling or substituting the map multiplies  $\sigma_J$  by 2.18 and 2.13 (to 47.0 and 46.0 vs. 21.6 baseline) while a symmetric category swap leaves it unchanged (Figure 2(h); Table 11). The attribution *target* thus changes with the assumed inventory even when recovery is exact. The tables for coherence and grouped reporting, missing-source adequacy and per-sensor footprints are in Appendix G.

Table 6: Experiment 1: coefficient recovery versus conditioning and noise. The better-conditioned geometry recovers more accurately at matched noise.

| geom.     | noise | $\sigma_J$ | $\kappa$ | coef. err. | resid. |
|-----------|-------|------------|----------|------------|--------|
| separated | 0.00  | 3.27       | 8.89     | 0.000      | 0.00   |
| separated | 0.05  | 3.27       | 8.89     | 0.066      | 6.37   |
| separated | 0.10  | 3.27       | 8.89     | 0.132      | 12.73  |
| separated | 0.20  | 3.27       | 8.89     | 0.264      | 25.46  |
| close     | 0.00  | 9.60       | 2.26     | 0.000      | 0.00   |
| close     | 0.05  | 9.60       | 2.26     | 0.028      | 8.42   |
| close     | 0.10  | 9.60       | 2.26     | 0.057      | 16.84  |
| close     | 0.20  | 9.60       | 2.26     | 0.113      | 33.68  |

## F EXPANDED RESULTS

Full numeric tables backing Figures 2 and 3.

**Summary of further controlled results.** From  $L = 4$  to 16 hours  $\sigma_J$  rises from 0.08 to 9.07, but the operator still changes by 6–14% between adjacent lags (Experiment 7); activity trajectories have lower error than coefficients (Experiment 9);  $\sigma_J$  moves from 21.6 to 47.0 under a rescaled inventory map although recovery stays exact (Experiment 6); and a shifted copy of one map raises the coherence to 0.958 (separation 0.286) without a merge (Experiment 2) (Figure 2(f)–(h), Appendix F). Eight downwind sensors give coherence 0.9897 and 0.9901 under constant and single-direction wind (Table 7), on either side of 0.99, so the rule keeps the two sources apart in the first case and merges them in the second (the run records coherences, not groupings).

### F.1 Conditioning (Exp 1)

Coefficient recovery versus conditioning and noise; the better-conditioned geometry recovers more accurately at matched noise.

### F.2 Wind Diversity and Sensor Geometry (Exp 4)

$\sigma_J$ , maximum eligible coherence, and coefficient error across wind provider  $\times$  sensor layout, and the wind-window ensemble spread. With the eight randomly placed sensors and the constant, single, diurnal and AR(1) winds, the operator column of source B has norm  $5.2 \times 10^{-7}$ ,  $1.7 \times 10^{-4}$ ,  $5.2 \times 10^{-5}$  and  $1.1 \times 10^{-4}$ , against  $7.4 \times 10^{-4}$ , 0.44, 0.065 and 0.22 for source A and 7.6–18.6 for both sources under the regulatory layout (recomputed from the same configuration). In these cells  $\sigma_J$  is near zero because source B is nearly invisible, and the coherence is computed on a column that carries almost no signal.

### F.3 Background Stress (Exp 3)

Redundant  $\equiv$  primary (same span); the source-like stress basis drives  $\sigma_J$  to 0 and absorption to 1.

### F.4 Transport Error (Exp 5)

Parametric perturbations (wind speed, direction, dispersion); a structural PDE mismatch generated with the auxiliary edge-hold advection–diffusion solver yields adequacy rejection rate 1.0.

### F.5 Inventory Robustness (Exp 6)

Location, scale, category, and map-version perturbations, each a separate scenario never pooled with transport error; recovery exact,  $\sigma_J$  tracks the version.

### F.6 Lag-Window Selection (Exp 7)

Conditioning improves with lag. No candidate meets the convergence threshold  $\tau_L = 10^{-3}$  of Equation 15; the code then records the largest candidate ( $L = 16$ ) as a fallback. The other reported runs use the declared  $L = 12$ .

Table 7: Experiment 4:  $\sigma_J$ , maximum eligible coherence, and coefficient error across wind provider  $\times$  sensor layout.

| wind     | layout     | $\sigma_J$ | max. coh. | coef. err. |
|----------|------------|------------|-----------|------------|
| constant | regulatory | 1.53       | 0.975     | 0.000      |
| constant | random     | 0.00       | 0.028     | 0.000      |
| constant | downwind   | 0.13       | 0.9897    | 0.000      |
| single   | regulatory | 6.44       | 0.741     | 0.000      |
| single   | random     | 0.00       | 1.000     | 0.573      |
| single   | downwind   | 1.36       | 0.9901    | 0.000      |
| diurnal  | regulatory | 6.11       | 0.722     | 0.000      |
| diurnal  | random     | 0.00       | 0.880     | 0.001      |
| diurnal  | downwind   | 2.60       | 0.897     | 0.000      |
| ar1      | regulatory | 7.35       | 0.641     | 0.000      |
| ar1      | random     | 0.00       | 0.774     | 0.115      |
| ar1      | downwind   | 2.14       | 0.916     | 0.000      |
| multi    | regulatory | 5.70       | 0.129     | 0.000      |
| multi    | random     | 2.79       | 0.322     | 0.000      |
| multi    | downwind   | 0.41       | 0.110     | 0.000      |
| real     | regulatory | 2.85       | 0.156     | 0.000      |
| real     | random     | 2.10       | 0.287     | 0.000      |
| real     | downwind   | 0.38       | 0.221     | 0.000      |

Table 8: Experiment 4: wind-window ensemble distribution ( $\sigma_J$  percentiles; both ensembles reach full rank with probability 1 and show no ambiguity).

| ensemble                 | $\sigma_J^5$ | $\sigma_J^{50}$ | $\sigma_J^{95}$ |
|--------------------------|--------------|-----------------|-----------------|
| historical (real slices) | 1.45         | 4.51            | 9.96            |
| simulated (AR(1))        | 3.50         | 5.58            | 6.80            |

## F.7 Temporal-Basis Recovery (Exp 9)

Activity-trajectory error stays below coefficient error at every noise level (fixed  $\sigma_J$ ), a fixed geometric property of the temporal-basis map.

## F.8 Baseline Comparison (Exp 11)

IASA against plain NNLS on the projected system (the same estimator without grouping or diagnostics), chemical mass balance (NNLS on the unprojected response) and a PMF/NMF receptor factorization, on the steady-wind case (one wind direction, six random sensors) and the source-like background, seeds 0–2, run on CPU. Shares are  $L_1$  fractions of the projected group signals  $\tilde{A}_k \hat{\mathbf{c}}_k$ , the quantity the signal target identifies; the group-signal error is  $\|\tilde{A}_G(\hat{\mathbf{c}}_G - \mathbf{c}_G)\|_2 / \|\tilde{A}_G \mathbf{c}_G\|_2$  for the blocks of IASA’s reported partition, and the table gives the largest over blocks. PMF/NMF has no coefficients in the operator’s units, so only its share error on the unprojected response is given. In seeds 0 and 1 of the steady-wind case both operator columns have norm below  $10^{-10}$  (against 0.06 in seed 2): no signal reaches the six sensors, IASA flags  $\sigma_J \approx 0$ , and no error is reported. Shares computed from the unprojected response include the absorbed column of the background-stress scenario, whose coefficient is not identified: plain NNLS assigns it values above  $10^{13}$  in seeds 1 and 2, which gives unprojected share errors of 1.30 and 1.26 although its group-signal errors are within 0.003 of IASA’s.

## F.9 Observed Weeks: Full Analysis

**Data and windows.** We use hourly government PM<sub>2.5</sub> and wind records (CPCB, 2023, 2025) from the 32-sensor New Delhi regulatory network, four proxy source groups (brick kilns, industries, population density, traffic) with declared temporal bases, and the gridded wind field of Appendix D. After the fixed-zero set,  $J = 7$ : one temporal component each for brick kilns, industries and population, and four time-of-day components for traffic. The declared partition  $\mathcal{D}$  has one block per source group. Four consecutive one-week windows, 1–28 May 2018, are fitted and reported separately, each on the 79–85% of sensor-hours with a valid reading ( $N = 4,236\text{--}4,561$ ). There

Table 9: Experiment 3: background stress. Redundant  $\equiv$  primary (same span); the source-like stress basis drives  $\sigma_J$  to 0 and absorption to 1.

| background | min. vis. | absorp. | $\sigma_J$ | coef. err. |
|------------|-----------|---------|------------|------------|
| none       | 1.000     | 0.000   | 0.080      | 2.36       |
| primary    | 0.972     | 0.235   | 0.080      | 2.61       |
| redundant  | 0.972     | 0.235   | 0.080      | 2.61       |
| stress     | 0.000     | 1.000   | 0.000      | 0.78       |

Table 10: Experiment 5: parametric transport error. Coefficient error rises with the operator error norm on the direction and dispersion axes (the speed axis is non-monotonic). Structural PDE mismatch: adequacy rejection rate 1.0.

| axis                    | value | op. err. | $\sigma_J$ | coef. err. |
|-------------------------|-------|----------|------------|------------|
| direction ( $^\circ$ )  | 0     | 0.00     | 0.87       | 0.00       |
| direction ( $^\circ$ )  | 5     | 0.29     | 3.37       | 0.62       |
| direction ( $^\circ$ )  | 10    | 0.60     | 7.07       | 0.80       |
| direction ( $^\circ$ )  | 20    | 0.87     | 7.98       | 0.86       |
| speed ( $\times$ )      | 1.25  | 0.29     | 2.09       | 0.45       |
| speed ( $\times$ )      | 1.50  | 0.53     | 4.57       | 0.27       |
| dispersion ( $\times$ ) | 1.50  | 0.19     | 1.42       | 0.50       |
| dispersion ( $\times$ ) | 2.00  | 0.35     | 1.76       | 0.89       |

is no ground truth for source contributions.

**Noise level.** The two co-located Pusa monitors differ with standard deviation  $43.4 \mu\text{g}/\text{m}^3$  over 17,949 shared hours (none in May 2018), so if their errors are independent with equal variance each has  $\sigma_e = 30.7 \mu\text{g}/\text{m}^3$ . The difference has lag-1 autocorrelation 0.857; for errors independent across sensors and AR(1) in time with this coefficient, the error covariance has spectral norm at most  $(1 + 0.857)/(1 - 0.857) = 13.0$  times  $\sigma_e^2$ , an effective noise level of  $110.7 \mu\text{g}/\text{m}^3$ . Both describe instrument error only; operator error adds to them (Proposition D.1).

**Results.** Section 6.2 gives the identifiable resolution, the group signals against the noise level and the shares with their intervals. The tables below give the per-week identifiability, the residual diagnostics and, for each source group, its separation (the sine of the smallest angle between the span of the group’s fingerprints and the span of the other fingerprints) and the root mean square of its fitted signal per reading. The fit subtracts each week’s transported initial-condition baseline, and the residual adequacy test is reported as uncalibrated because no calibrated noise model for the observed  $\text{PM}_{2.5}$  is available.

## F.10 Observed New Delhi (weeks 1–4)

Per-week identifiability geometry and sensor-fit/residual diagnostics ( $\text{PM}_{2.5}$  never imputed; adequacy uncalibrated).

Table 18 gives the signal-target analysis of Section 6.2, computed from the committed projected operators and fits of the four weeks with `experiments/iasa_pol/signal_target_weeks.py`. The bootstrap adds Gaussian noise in  $\text{col}(\tilde{A})$  to the fitted signal, which is the only part of the noise that changes the NNLS fit, and refits with an active-set NNLS solver (`scipy.optimize.nnls`), whose solution agrees with the projected-FISTA fit to five decimals on the four weeks. The reported groupings in this table and in the Experiment 11 scoring were computed with a separation rule (every block must have separation at least  $\sqrt{1 - 0.99^2} = 0.141$ ); on these operators it gives the same blocks as the pairwise coherence rule.

## G ADDITIONAL EXPERIMENTS

Three further controlled experiments not shown in the main-text figures.

Table 11: Experiment 6: inventory robustness on the real maps. Recovery is exact in every scenario;  $\sigma_J$  tracks the inventory version.

| scenario         | $\sigma_J$ | coef. err. |
|------------------|------------|------------|
| baseline         | 21.56      | 0.0        |
| location shift   | 28.85      | 0.0        |
| spatial rescale  | 47.00      | 0.0        |
| alt. map version | 46.01      | 0.0        |
| category swap    | 21.56      | 0.0        |

Table 12: Experiment 7: lag window. Conditioning improves with lag; no candidate meets  $\tau_L$ , and the other reported runs use the declared  $L = 12$ .

| lag $L$ | $\sigma_J$ | $\kappa$ | #comp. | coef. err. |
|---------|------------|----------|--------|------------|
| 4       | 0.08       | 569.2    | 2      | 0.0        |
| 6       | 0.71       | 63.8     | 2      | 0.0        |
| 8       | 2.46       | 18.6     | 2      | 0.0        |
| 10      | 4.57       | 10.1     | 2      | 0.0        |
| 12      | 6.16       | 7.5      | 2      | 0.0        |
| 16      | 9.07       | 5.1      | 2      | 0.0        |

### G.1 Coherence and Grouped Reporting (Exp 2)

We form increasingly coherent source pairs by shifting a copy of one inventory map by a controlled spatial offset, so decreasing the shift drives coherence toward one. Shrinking the offset drives maximum eligible coherence from 0.05 to 0.96 while  $\sigma_J$  falls from 21.8 to 4.6, yet at every offset the two sources stayed separate groups with exact recovery (individual relative error 0; Table 20). The largest coherence, 0.958, is below  $\tau_\rho = 0.99$ , so the pairwise rule keeps the two sources apart.

### G.2 Missing-Source Adequacy (Exp 8)

On the real New Delhi platform with a non-empty rank-4 primary background, the refitted parametric bootstrap ( $\alpha = 0.05$ , 1000 replicates) rejects at rate 0.05 under the null and at rate 1.0 against a residual-visible omitted source whose signature lies largely outside  $\text{span}[H_\Phi^{\text{lag}}, Q]$  (rejection 1.0); an *aligned* omission, a nonnegative combination of the fitted columns and background, is absorbed and not detected (rejection 0.05; Table 21). A rejection is therefore evidence of model inadequacy, whereas a non-rejection does not show that the inventory is complete.

### G.3 Per-Sensor Footprints (Exp 10)

On controlled trials the per-sensor footprints of Appendix D.5 localize the responsible upwind cells (mass centroid 0.74 cells from the true origin, 100% of mass within the 4-cell radius, nonnegative) and per-sensor contributions sum to the fitted sensor signal to machine precision; because a non-singleton component is present, footprints are reported at group granularity. On observed New Delhi data footprints are reported per monitor at the identifiable report groups, accompanying the apportionment of Section 6.2 (Appendix D.5).

## H EXTENDED DISCUSSION ON RELATED WORK

This appendix extends the related work of Section 2.

**Estimability and identification.** In the linear model  $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$ , a linear function  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is estimable exactly when  $\boldsymbol{\lambda}$  lies in the row space of  $X$  (Searle, 1971). In general parametric models, local identification holds at a regular point exactly when the Fisher information matrix is nonsingular (Rothenberg, 1971). Our rank condition is the special case of both for model (2) after nuisance projection, and identifiability of a coefficient sum is estimability of an indicator functional. When point identification fails, partial identification reports the set of parameter values consistent with the data (Manski, 2003). Our question is which groupings of declared,

Table 13: Experiment 9: temporal-basis recovery (bases: diurnal, block, day/night). Activity-trajectory error stays below coefficient error at every noise level.

| noise | coef. err. | activity err. |
|-------|------------|---------------|
| 0.00  | 0.000      | 0.000         |
| 0.02  | 0.132      | 0.075         |
| 0.05  | 0.329      | 0.188         |
| 0.10  | 0.659      | 0.375         |
| 0.20  | 0.900      | 0.582         |

Table 14: Experiment 11, seeds 0, 1, 2: projected share error ( $L_2$ ) and largest relative group-signal error. In the steady-wind case only seed 2 has signal at the sensors; IASA merges its two sources and reports the merged group. <sup>†</sup>Share error on the unprojected response.

| scenario           | method     | flag | share err.                    | group-signal err.   |
|--------------------|------------|------|-------------------------------|---------------------|
| bg. stress         | IASA       | yes  | 0.007, 0.003, 0.007           | 0.054, 0.028, 0.060 |
| bg. stress         | plain NNLS | no   | 0.007, 0.003, 0.006           | 0.054, 0.028, 0.062 |
| bg. stress         | CMB        | no   | 0.193, 0.123, 0.190           | 4.63, 2.10, 4.32    |
| bg. stress         | PMF/NMF    | no   | 1.15, 1.19, 1.18 <sup>†</sup> | –                   |
| steady wind (s. 2) | IASA       | yes  | merged                        | 0.004               |
| steady wind (s. 2) | plain NNLS | no   | 0.036                         | 0.004               |
| steady wind (s. 2) | CMB        | no   | 0.558                         | 7.84                |
| steady wind (s. 2) | PMF/NMF    | no   | 0.304 <sup>†</sup>            | –                   |

nonnegative source components can be reported, with a nuisance subspace and an operator that is itself estimated.

**Collinearity and matroids.** Regression diagnostics measure how far each regressor, or block of regressors, is from the span of the others: the variance-inflation factor for one column, the generalized variance-inflation factor of Fox and Monette (1992) for a block, which combines all canonical correlations between the block and the rest, and the condition indices and variance-decomposition proportions of Belsley et al. (1980), which identify the variables involved in each near dependency. None of them returns a partition. The connected components of a matroid can be read off the fundamental circuits of any basis (Kroghdahl, 1977; Oxley, 2011); we use this fact to characterize which partitions of declared components have identifiable group signals.

**Subspace clustering.** Sparse subspace clustering groups data points that lie in a union of subspaces; when the subspaces are independent, meaning the dimension of their sum equals the sum of their dimensions (our direct-sum condition), each point has a sparse representation that uses only points from its own subspace (Elhamifar and Vidal, 2013), and geometric and noisy analyses give recovery conditions in terms of subspace angles and noise (Soltanolkotabi and Candès, 2012; Wang and Xu, 2016). Theorem 5.7(c) is a related statement about the columns of an operator. It differs in three respects: we decompose a given operator and do not assume a union-of-subspaces model with points in general position; the decomposition serves a regression target, each group’s contribution to the projected signal; and the operator is formed after nuisance projection and has nonnegative coefficients.

**Air-pollution source apportionment.** Receptor-oriented methods estimate source contributions from ambient chemical or physical measurements, by chemical mass balance or by factor analysis such as positive matrix factorization (Watson et al., 2002; Paatero and Tapper, 1994; Hopke, 2016); source-oriented methods propagate emission inventories through atmospheric transport. Reviews and harmonization efforts describe practice for both and document sensitivity to source profiles, factor interpretation, chemical resolution and model comparability (Viana et al., 2008; Belis et al., 2013, 2019; Mircea et al., 2020; Hopke et al., 2020). In India, studies of the same city report divergent source contributions because of limited local profiles, collinear tracers and differences in method (Pant and Harrison, 2012), and global reviews compile source categories such as traffic, industry, dust and domestic combustion from heterogeneous studies (Karagulian et al., 2015). For CMB with nearly collinear profiles, Henry (1992) defines an eligible space of combinations of contributions that can be estimated to a stated precision; for PMF, rotational ambiguity is explored with dedicated tools (Paatero et al., 2002). In statistics, Park et al. (2001) fit multivariate receptor models with temporally correlated data by Markov chain Monte Carlo, and Jin and Datta (2025) give identification conditions and consistent estimation in source apportionment. These methods use multi-species chemical measurements. Our setting has one measured species ( $\text{PM}_{2.5}$  mass) at a

Table 15: Observed New Delhi identifiability, weeks 1–4. All weeks are full rank, finite-conditioned, reported as the four source groups (traffic has four columns).

| week | $\sigma_1$ | $\sigma_J$ | num. rank | max. coh. | groups   |
|------|------------|------------|-----------|-----------|----------|
| 1    | 50.1       | 3.71       | 7         | 0.316     | 4 groups |
| 2    | 60.3       | 5.67       | 7         | 0.355     | 4 groups |
| 3    | 77.0       | 10.15      | 7         | 0.496     | 4 groups |
| 4    | 81.2       | 7.65       | 7         | 0.475     | 4 groups |

Table 16: Observed New Delhi sensor fit and residual diagnostics, weeks 1–4. PM<sub>2.5</sub> never imputed; adequacy uncalibrated (no verdict).

| week | obs. rows | mask frac. | resid. norm | adequacy     |
|------|-----------|------------|-------------|--------------|
| 1    | 4561      | 0.848      | 3768        | uncalibrated |
| 2    | 4278      | 0.796      | 5955        | uncalibrated |
| 3    | 4315      | 0.803      | 3050        | uncalibrated |
| 4    | 4236      | 0.788      | 3760        | uncalibrated |

sparse network, and separability comes from wind-conditioned transport and timing; we do not propose another receptor model or inventory, but ask when proxy source maps can be distinguished at the sensors after transport, lags, nuisance removal and noise.

**Inverse modelling and physics-constrained learning.** Estimating latent states or parameters from partial observations is the subject of inverse problems and data assimilation (Tarantola, 2005; Engl et al., 1996); Kalman filtering, ensemble Kalman filters and variational assimilation combine a dynamical model with priors, covariances and an observation model (Wang et al., 2023; Carrassi et al., 2018). Atmospheric inversion measures which elements of the state the observations constrain through the averaging kernel and the degrees of freedom for signal (Rodgers, 2000), and chooses how far to aggregate state-vector elements by balancing aggregation error against smoothing error (Turner and Jacob, 2015). Physics-informed and operator-learning methods (Raissi et al., 2019; Li et al., 2024, 2020; Bhardwaj et al., 2025b) add governing equations or physical structure to flexible models. These methods regularize or aggregate with a prior; IASA reports the grouping that the projected operator supports, with every threshold declared before fitting.

**Sensor-space spatio-temporal learning.** Graph neural networks, transformers and sequence models predict spatio-temporal processes from sensor data, including traffic and air quality (Li et al., 2017; Yu et al., 2017; Wu et al., 2019; Chen et al., 2023; Iyer et al., 2022; Bhardwaj et al., 2025a,b). Their objective is forecasting, interpolation or representation learning, not attribution to declared source inventories; a model can predict the readings well while the source groups remain indistinguishable at the deployed sensors.

**Observability and system identification.** Whether latent quantities can be recovered from outputs is the observability question of control theory (Chen, 1984) and of system identification and inverse problems for nonlinear systems and PDEs (Ljung, 1987; Banks and Kunisch, 2012; Colton et al., 1990). We ask it for declared source components after wind-conditioned transport, finite lags, nuisance removal and noise, and report the answer with the estimates.

## I IMPLEMENTATION INVARIANTS

The response is constructed first on the complete time-major, sensor-minor row index. The PM<sub>2.5</sub> observation mask then selects the same ordered rows from  $Y$ ,  $H_{\Phi}^{\text{lag}}$ ,  $Q$ , and metadata. Source-basis columns are always source-major and basis-minor. Any row or column metadata mismatch is a hard error.

The response construction uses integer source-cell centers, half-cell domain extents, fractional final advection substeps, a positive minimum dispersion age, direct sensor evaluation, finite lag, and complete removal after center exit. Kernel boundary/truncation loss and whole-release exit loss are recorded separately. Retained and dropped kernel mass sum to emitted kernel mass within the recorded quadrature tolerance; sensor observations are never summed as a mass proxy. The default puff baseline has zero source and zero initial state.



Table 17: Observed weeks, largest group: share of the fitted projected signal, 95% bootstrap interval and  $b_g/\|\tilde{A}_g\hat{c}_g\|_2$  at the instrument noise level 30.7 and the AR(1) upper level 110.7  $\mu\text{g}/\text{m}^3$ . All groups: Table 19. The bootstrap draws independent noise at each level; its percentile intervals need not contain the estimate, because NNLS sets some coefficients to zero.

| wk | group       | share | interval  |           | bound/norm |       |
|----|-------------|-------|-----------|-----------|------------|-------|
|    |             |       | 30.7      | 110.7     | 30.7       | 110.7 |
| 1  | population  | 1.00  | 0.75–1.00 | 0.26–0.99 | 0.49       | 1.78  |
| 2  | population  | 0.81  | 0.71–0.85 | 0.49–0.93 | 0.24       | 0.88  |
| 3  | population  | 0.94  | 0.00–0.94 | 0.00–0.92 | 2.96       | 10.67 |
| 4  | brick kilns | 0.93  | 0.33–1.00 | 0.00–1.00 | 1.33       | 4.81  |

Table 18: Observed New Delhi on the signal target.  $s_g$ : separation of the source group; signal: root mean square of the fitted projected group signal per reading ( $\mu\text{g}/\text{m}^3$ ); share with 95% parametric-bootstrap interval at  $\sigma_e = 30.7$  (1000 refits).

| wk | group       | $s_g$ | signal | share | interval  |
|----|-------------|-------|--------|-------|-----------|
| 1  | brick kilns | 0.89  | 0.00   | 0.00  | 0.00–0.09 |
|    | industries  | 0.95  | 0.00   | 0.00  | 0.00–0.14 |
|    | population  | 0.87  | 5.40   | 1.00  | 0.75–1.00 |
|    | traffic     | 0.82  | 0.00   | 0.00  | 0.00–0.17 |
| 2  | brick kilns | 0.94  | 0.00   | 0.00  | 0.00–0.04 |
|    | industries  | 0.93  | 2.51   | 0.16  | 0.10–0.21 |
|    | population  | 0.83  | 11.86  | 0.81  | 0.71–0.85 |
|    | traffic     | 0.84  | 0.96   | 0.04  | 0.01–0.10 |
| 3  | brick kilns | 0.96  | 0.09   | 0.06  | 0.00–0.45 |
|    | industries  | 0.86  | 0.00   | 0.00  | 0.00–0.58 |
|    | population  | 0.78  | 1.03   | 0.94  | 0.00–0.94 |
|    | traffic     | 0.87  | 0.00   | 0.00  | 0.00–0.88 |
| 4  | brick kilns | 0.96  | 1.87   | 0.93  | 0.33–1.00 |
|    | industries  | 0.87  | 0.08   | 0.07  | 0.00–0.41 |
|    | population  | 0.72  | 0.00   | 0.00  | 0.00–0.32 |
|    | traffic     | 0.77  | 0.00   | 0.00  | 0.00–0.55 |

Normal backgrounds have effective rank at most eight and do not use inventory or response columns. Thin-SVD projection stores the numerical tolerance, singular values, retained  $U_r$ , dependent input columns, removed components, and orthogonality/idempotence residuals. A source-like column is permitted only when the run is labeled as a stress test.

The primary  $Q$ , lag-candidate grid, and fixed-zero coefficient set are declared before fitting. A fixed-zero mask removes columns before diagnostics and fitting and stores both reduced-to-original and original-to-reduced index maps. Fitted near-zero coefficients never modify the mask. Adjacent lag responses use identical rows and columns; the default convergence threshold is  $\tau_L = 10^{-3}$ . The reported runs use the declared  $L = 12$ ; the lag sweep of Experiment 7 records the largest candidate as its “selected” lag when no candidate meets  $\tau_L$ .

## J SANITY GATES

Before paper experiments, the implementation must pass six small deterministic gates built from public APIs:

1. **S1: response.** On a  $16 \times 16$  grid with four compact sources, four sensors, impulse/constant bases, and seed 123, verify response shape, downwind arrival, open-boundary exit, no wraparound, dispersion anisotropy, mass accounting, metadata, and exact repeatability.
2. **S2: projection.** Use the S1 response with a rank-four normal background to verify exact ordering, background removal, orthogonality, idempotence, redundant-column invariance, and the labeled source-like stress behavior.
3. **S3: diagnostics.** Verify orthogonal, duplicate, and weak matrices, including padded zeros,  $\sigma_J$ , infinite

Table 19: Observed weeks, all groups: share of the fitted projected signal and 95% parametric-bootstrap intervals (1,000 refits) at the instrument noise level  $\sigma_e = 30.7$  and at the AR(1) upper level 110.7 (both in  $\mu\text{g}/\text{m}^3$ ); the bootstrap draws independent noise at each level.

| wk | group       | share | at 30.7   | at 110.7  |
|----|-------------|-------|-----------|-----------|
| 1  | brick kilns | 0.00  | 0.00–0.09 | 0.00–0.26 |
|    | industries  | 0.00  | 0.00–0.14 | 0.00–0.45 |
|    | population  | 1.00  | 0.75–1.00 | 0.26–0.99 |
|    | traffic     | 0.00  | 0.00–0.17 | 0.00–0.51 |
| 2  | brick kilns | 0.00  | 0.00–0.04 | 0.00–0.11 |
|    | industries  | 0.16  | 0.10–0.21 | 0.00–0.33 |
|    | population  | 0.81  | 0.71–0.85 | 0.49–0.93 |
|    | traffic     | 0.04  | 0.01–0.10 | 0.00–0.27 |
| 3  | brick kilns | 0.06  | 0.00–0.45 | 0.00–0.67 |
|    | industries  | 0.00  | 0.00–0.58 | 0.00–0.86 |
|    | population  | 0.94  | 0.00–0.94 | 0.00–0.92 |
|    | traffic     | 0.00  | 0.00–0.88 | 0.00–1.00 |
| 4  | brick kilns | 0.93  | 0.33–1.00 | 0.00–1.00 |
|    | industries  | 0.07  | 0.00–0.41 | 0.00–0.74 |
|    | population  | 0.00  | 0.00–0.32 | 0.00–0.61 |
|    | traffic     | 0.00  | 0.00–0.55 | 0.00–1.00 |

Table 20: Experiment 2: coherence rises as the source offset shrinks; no pair exceeds  $\tau_\rho = 0.99$  and noiseless recovery stays exact.

| offset | max. coh. | $\sigma_J$ | merged | indiv./grp. err. |
|--------|-----------|------------|--------|------------------|
| 8.0    | 0.048     | 21.77      | no     | 0.0 / 0.0        |
| 6.0    | 0.862     | 10.48      | no     | 0.0 / 0.0        |
| 4.0    | 0.905     | 8.40       | no     | 0.0 / 0.0        |
| 2.0    | 0.945     | 5.43       | no     | 0.0 / 0.0        |
| 1.0    | 0.958     | 4.64       | no     | 0.0 / 0.0        |

deficient condition status, weak-pair N/A values, and a matched multi-source eastward versus two-direction wind comparison.

4. **S4: fit.** Recover known nonnegative coefficients in noiseless/noisy well-conditioned cases, verify projected-FISTA KKT convergence, preserve exact nonnegativity, and recover the duplicate-pair sum when the individual split is nonunique.
5. **S5: report groups.** Verify the pairwise report-group rule used in the recorded controlled runs: the duplicate-source connected component, an  $A$ – $B$ – $C$  chain with both trigger edges retained, the separated-source non-edge, the independent weak flag, and the summed group contribution.
6. **S6: end to end.** Build sources, wind, response, background, masked observations, diagnostics, fit, uncertainty, and report groups with one command and write a compact provenance-complete summary.

The experiments begin only after S1–S6 and the PyTorch CPU/CUDA parity checks pass. **S8 (grouping)** covers Theorem 5.7: the matroid components equal the unique finest identifiable partition found by brute force on random operators, a zero column is its own component, and the declared partition is respected. Two further gates support the results. **S7 (adequacy calibration)** runs a statistically calibrated study of the residual-adequacy test on a controlled orthonormal design, checking that the null rejection rate matches  $\alpha$ , that the null  $p$ -values are uniform, and that power increases monotonically with omission strength. A final **reporting gate** exercises the evaluation layer that emits the result tables in Appendix E, verifying our reporting rules: undefined weak-pair metrics stay null, grouped metrics accompany any non-singleton report component, percentages are labeled fractions of fitted sensor signal, and an uncalibrated run never reports an adequacy pass.

Table 21: Experiment 8: missing-source adequacy on the platform (non-empty  $Q$ ). Calibrated null, full power on a residual-visible omission, and no detection of an absorbed in-span omission.

| omission case                              | rejection rate |
|--------------------------------------------|----------------|
| none (null, calibration)                   | 0.05           |
| residual-visible (out of span, frac. 0.91) | 1.00           |
| aligned (in span, absorbed)                | 0.05           |

## K PYTORCH SOLVER AND UNCERTAINTY PROTOCOL

After CSV/NPZ ingestion, numerical computation uses PyTorch tensors on an explicit device and dtype. The nonnegative coefficient objective is

$$f(\mathbf{c}) = \|\tilde{Y} - \tilde{H}_{\Phi}\mathbf{c}\|_2^2 + \lambda\|\mathbf{c} - \mathbf{c}_0\|_2^2, \quad \mathbf{c} \geq 0. \quad (27)$$

Every reported run uses  $\lambda = 0$ , which is problem (4). Projected FISTA uses a Lipschitz step derived from  $\sigma_1$  or a recorded power-iteration estimate, clamps every iterate to the nonnegative orthant, and restarts acceleration whenever the objective increases. Termination requires the configured projected-gradient/KKT residual and relative objective-change tolerances, or reports that the iteration cap was reached. The fit artifact stores convergence status, iteration count, final KKT residual, objective summary, device, and dtype.

Active-set covariance uses PyTorch linear algebra on  $\tilde{H}_{\Phi,\mathcal{A}}^\top \tilde{H}_{\Phi,\mathcal{A}} + \lambda I$ . Wind-ensemble and bootstrap refits are batched on the same device when shapes permit. Undefined weak-pair metrics are serialized as JSON `null`, never NaN or a false numerical distance.

Uncertainty artifacts distinguish `ensemble_kind=transport` from `ensemble_kind=inventory`. Only the former may be summarized as an ensemble uncertainty interval; inventory members remain named robustness scenarios, and an aggregation request that mixes the two kinds is an error.

## L ADEQUACY AND ENSEMBLE RESULT CONTRACTS

A residual-adequacy result stores `calibration_status`,  $T_{\text{res}}$ , bootstrap quantile, Monte Carlo  $p$ -value,  $\alpha$ , replicate count, and noise-model provenance. Paper runs use  $\alpha = 0.05$  and 1000 refitted replicates, except the structural-mismatch case of Experiment 5, which uses 200. If the externally calibrated noise covariance is absent or invalid, numerical test fields are null and the status is `uncalibrated`; no boolean pass is emitted. Sensor, time, and autocorrelation residual summaries are retained in either case.

A lag-selection result stores every candidate  $L$ , adjacent convergence ratio, selected  $L$ ,  $\tau_L$ , physical grid rationale, row count, and diagnostic/group summaries. A wind-distribution result stores the sampled-window provenance; 5th, 50th, and 95th percentiles of  $\sigma_J$ ; full numerical and effective-rank probabilities; coefficient weakness probabilities; separation distributions; and reported-partition frequencies.

## M REPRODUCIBILITY AND ARTIFACT PROVENANCE

Every paper run records the container image, repository revision, PyTorch and CUDA versions, device model, dtype, deterministic-algorithm settings, random seeds, source files and crop, per-source scales, wind-imputer architecture, held-out wind split, query grid, gridded wind product, and ensemble provenance, inventory identifier/hash/version, observation mask, response and transport configuration, wind realization, dispersion settings, background columns and rank tolerance, lag protocol, fixed-zero mask and index mapping, diagnostic thresholds, ensemble kind, solver settings, noise-model provenance, and parent artifact hashes. Generated wind products, response matrices, checkpoints, and experiment outputs are written to generated-data or log paths and are not treated as immutable source files.

Controlled-run artifacts contain true and estimated coefficients, reconstructed activities, response/projection results, diagnostics, uncertainty, report groups, and recovery metrics. Observed New Delhi artifacts omit unavailable ground truth and instead contain observed-row counts, fitted trajectories, raw and projected residuals, normalized proxy activities, group signals, uncertainty, weak/ambiguous flags, trigger pairs, and group summaries.

A percentage is emitted only with an explicit denominator identifying it as a fraction of fitted inventory-attributed sensor signal.

**Computing infrastructure.** The paper-resolution controlled experiments (Experiments 1–10) and the observed fits ran on one NVIDIA GPU with CUDA 11.8 and PyTorch 2.6; the response operator is stored in float32. Experiments 1–10 were re-run from the recorded configurations on one laptop CPU (Apple M5, PyTorch 2.14), and every recorded scalar result agrees with the GPU run to within 1% relative difference. Experiment 11 and the signal-target re-analysis of the observed weeks ran on that CPU; the re-analysis reads the committed projected operators and fits and does not re-run the pipeline.

**Data and licences.** The  $\text{PM}_{2.5}$  and meteorological records are public data of the Central Pollution Control Board of India (CPCB, 2023, 2025). The population grid is GPWv4, Revision 11 (Center for International Earth Science Information Network (CIESIN), Columbia University, 2018), distributed under the Creative Commons Attribution 4.0 licence. The brick-kiln and industry maps follow the regional emissions inventory of Guttikunda and Calori (2013), and the traffic map is derived from road map data (Google, 2024); Appendix D.6 describes the preparation. The code is in the supplementary material.